

Exercise Sheet 4: INTEGRALS

January 3, 2026

Riemann Sum

Question 1

Estimate the area under the graph of $f(x) = \frac{1}{x}$ from $x = 1$ to $x = 5$ using four approximating rectangles ($n = 4$) and taking the sample points to be: (a) Right endpoints (R_4) (b) Left endpoints (L_4)

Solution The interval is $[1, 5]$ and $n = 4$. The width of each subinterval is $\Delta x = \frac{5-1}{4} = 1$. The endpoints are $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 5$.

(a) **Right Endpoints (R_4)** We use $x_i = 2, 3, 4, 5$:

$$R_4 = \sum_{i=1}^4 f(x_i)\Delta x = f(2)(1) + f(3)(1) + f(4)(1) + f(5)(1)$$

$$R_4 = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{30 + 20 + 15 + 12}{60} = \frac{77}{60}$$

Since $f(x) = 1/x$ is a decreasing function, R_4 is an **underestimate**.

(b) **Left Endpoints (L_4)** We use $x_{i-1} = 1, 2, 3, 4$:

$$L_4 = \sum_{i=1}^4 f(x_{i-1})\Delta x = f(1)(1) + f(2)(1) + f(3)(1) + f(4)(1)$$

$$L_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12 + 6 + 4 + 3}{12} = \frac{25}{12}$$

Since $f(x) = 1/x$ is a decreasing function, L_4 is an **overestimate**.

Question 2

Estimate the area under the graph of $f(x) = 25 - x^2$ from $x = 0$ to $x = 5$ using five approximating rectangles ($n = 5$) and taking the sample points to be: (a) Right endpoints (R_5) (b) Left endpoints (L_5)

Solution The interval is $[0, 5]$ and $n = 5$. The width of each subinterval is $\Delta x = \frac{5-0}{5} = 1$. The endpoints are $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$.

(a) **Right Endpoints (R_5)** We use $x_i = 1, 2, 3, 4, 5$. The function values are: $f(1) = 24, f(2) = 21, f(3) = 16, f(4) = 9, f(5) = 0$.

$$R_5 = \sum_{i=1}^5 f(x_i)\Delta x = 1 \cdot (24 + 21 + 16 + 9 + 0) = \mathbf{70}$$

Since $f(x) = 25 - x^2$ is decreasing on $[0, 5]$, R_5 is an **underestimate**.

(b) **Left Endpoints (L_5)** We use $x_{i-1} = 0, 1, 2, 3, 4$. The function values are: $f(0) = 25, f(1) = 24, f(2) = 21, f(3) = 16, f(4) = 9$.

$$L_5 = \sum_{i=1}^5 f(x_{i-1})\Delta x = 1 \cdot (25 + 24 + 21 + 16 + 9) = \mathbf{95}$$

Since $f(x)$ is decreasing, L_5 is an **overestimate**.

Question 3

Estimate the area under the graph of $f(x) = 1 + x^2$ from $x = -1$ to $x = 2$ using three approximating rectangles ($n = 3$) and taking the sample points to be: (a) Right endpoints (R_3) (b) Left endpoints (L_3) (c) Midpoints (M_3)

Solution The interval is $[-1, 2]$ and $n = 3$. The width of each subinterval is $\Delta x = \frac{2 - (-1)}{3} = 1$. The endpoints are $x_0 = -1, x_1 = 0, x_2 = 1, x_3 = 2$.

(a) **Right Endpoints (R_3)** We use $x_i = 0, 1, 2$:

$$R_3 = f(0)(1) + f(1)(1) + f(2)(1) = (1 + 0^2) + (1 + 1^2) + (1 + 2^2) = 1 + 2 + 5 = 8$$

(b) **Left Endpoints (L_3)** We use $x_{i-1} = -1, 0, 1$:

$$L_3 = f(-1)(1) + f(0)(1) + f(1)(1) = (1 + (-1)^2) + (1 + 0^2) + (1 + 1^2) = 2 + 1 + 2 = 5$$

(c) **Midpoints (M_3)** The midpoints are $x_1^* = -0.5, x_2^* = 0.5, x_3^* = 1.5$. The function values are: $f(-0.5) = 1 + (-0.5)^2 = 1.25, f(0.5) = 1 + (0.5)^2 = 1.25, f(1.5) = 1 + (1.5)^2 = 3.25$.

$$M_3 = f(-0.5)(1) + f(0.5)(1) + f(1.5)(1) = 1.25 + 1.25 + 3.25 = 5.75$$

Question 4

Estimate the area under the graph of $f(x) = x^4$ from $x = 0$ to $x = 2$ as a limit. Do not evaluate the limit.

Solution The interval is $[a, b] = [0, 2]$. The width of each subinterval is $\Delta x = \frac{b-a}{n} = \frac{2}{n}$. The right endpoint of the i -th subinterval is $x_i = a + i\Delta x = 0 + i\left(\frac{2}{n}\right) = \frac{2i}{n}$. The area A is given by the limit of the Riemann sum R_n :

$$\begin{aligned} A &= \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\ A &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{2i}{n}\right)^4 \left(\frac{2}{n}\right) \\ A &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{16i^4}{n^4} \cdot \frac{2}{n}\right) = \lim_{n \rightarrow \infty} \frac{32}{n^5} \sum_{i=1}^n i^4 \end{aligned}$$

Question 5

Find a limit expression for the area under the curve $y = \frac{\ln x}{x}$ from $x = 1$ to $x = 3$ as a limit. Do not evaluate the limit.

Solution The interval is $[a, b] = [1, 3]$. The width of each subinterval is $\Delta x = \frac{3-1}{n} = \frac{2}{n}$. The right endpoint is $x_i = a + i\Delta x = 1 + \frac{2i}{n}$.

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\ln(1 + 2i/n)}{1 + 2i/n} \left(\frac{2}{n}\right)$$

Question 6

Find an expression for the area under the curve $y = e^x$ from $x = 1$ to $x = 3$ as a limit. Do not evaluate the limit.

Solution The interval is $[a, b] = [1, 3]$. The width of each subinterval is $\Delta x = \frac{3-1}{n} = \frac{2}{n}$. The right endpoint is $x_i = 1 + \frac{2i}{n}$.

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n e^{(1 + \frac{2i}{n})} \left(\frac{2}{n}\right)$$

Question 7

Determine a region whose area is equal to the limit:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left(5 + \frac{2i}{n} \right)^{10}$$

Solution Comparing this to $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$:

- $\Delta x = \frac{2}{n} \implies b - a = 2$.
- $x_i = 5 + \frac{2i}{n}$. Since $x_i = a + i\Delta x$, we have $a = 5$.
- $b = a + 2 = 5 + 2 = 7$.
- $f(x_i) = \left(5 + \frac{2i}{n} \right)^{10}$, and since $x_i = 5 + \frac{2i}{n}$, the function is $f(x) = x^{10}$.

The limit represents the area under the graph of $\mathbf{f}(\mathbf{x}) = \mathbf{x}^{10}$ from $\mathbf{x} = \mathbf{5}$ to $\mathbf{x} = \mathbf{7}$.

Question 8

Determine a region whose area is equal to the limit:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$$

Solution Comparing this to $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$:

- $\Delta x = \frac{3}{n} \implies b - a = 3$.
- $x_i = 1 + \frac{3i}{n}$. Since $x_i = a + i\Delta x$, we have $a = 1$.
- $b = a + 3 = 1 + 3 = 4$.
- $f(x_i) = \sqrt{1 + \frac{3i}{n}}$, and since $x_i = 1 + \frac{3i}{n}$, the function is $f(x) = \sqrt{x}$.

The limit represents the area under the graph of $\mathbf{f}(\mathbf{x}) = \sqrt{\mathbf{x}}$ from $\mathbf{x} = \mathbf{1}$ to $\mathbf{x} = \mathbf{4}$.

The Definite Integral

Question 1

Evaluate the integral $\int_{-1}^2 (1 + x^2) dx$.

Solution We use the Fundamental Theorem of Calculus, Part 2 (FTC2). An antiderivative of $f(x) = 1 + x^2$ is $F(x) = x + \frac{x^3}{3}$.

$$\begin{aligned} \int_{-1}^2 (1 + x^2) dx &= \left[x + \frac{x^3}{3} \right]_{-1}^2 \\ &= \left(2 + \frac{2^3}{3} \right) - \left(-1 + \frac{(-1)^3}{3} \right) \\ &= \left(2 + \frac{8}{3} \right) - \left(-1 - \frac{1}{3} \right) = \frac{14}{3} - \left(-\frac{4}{3} \right) = \frac{18}{3} = \mathbf{6} \end{aligned}$$

Question 2

Evaluate the integral $\int_0^3 (3 - 2x) dx$.

Solution An antiderivative of $f(x) = 3 - 2x$ is $F(x) = 3x - x^2$.

$$\begin{aligned} \int_0^3 (3 - 2x) dx &= [3x - x^2]_0^3 \\ &= (3(3) - 3^2) - (3(0) - 0^2) = (9 - 9) - 0 = \mathbf{0} \end{aligned}$$

Interpretation: The net area under the line $y = 3 - 2x$ from $x = 0$ to $x = 3$ is zero.

Question 3

Evaluate the integral $\int_{-2}^2 (3u+1)^2 du$.

Solution First, expand the integrand: $(3u+1)^2 = 9u^2 + 6u + 1$. An antiderivative is $F(u) = 3u^3 + 3u^2 + u$.

$$\begin{aligned}\int_{-2}^2 (9u^2 + 6u + 1) du &= [3u^3 + 3u^2 + u]_{-2}^2 \\ &= (3(2)^3 + 3(2)^2 + 2) - (3(-2)^3 + 3(-2)^2 + (-2)) \\ &= (24 + 12 + 2) - (-24 + 12 - 2) = 38 - (-14) = \mathbf{52}\end{aligned}$$

Question 4

Evaluate the integral $\int_0^4 (2v+5)(3v-1) dv$.

Solution First, expand the integrand: $(2v+5)(3v-1) = 6v^2 + 13v - 5$. An antiderivative is $F(v) = 2v^3 + \frac{13}{2}v^2 - 5v$.

$$\begin{aligned}\int_0^4 (6v^2 + 13v - 5) dv &= \left[2v^3 + \frac{13}{2}v^2 - 5v \right]_0^4 \\ &= \left(2(4)^3 + \frac{13}{2}(4)^2 - 5(4) \right) - (0) \\ &= (128 + 13(8) - 20) = 128 + 104 - 20 = \mathbf{212}\end{aligned}$$

Question 5

Evaluate the integral $\int_1^4 \frac{t^2+t}{t^{3/2}} dt$.

Solution Simplify the integrand: $\frac{t^2+t}{t^{3/2}} = t^{1/2} + t^{-1/2}$. An antiderivative is $F(t) = \frac{2}{3}t^{3/2} + 2t^{1/2}$.

$$\begin{aligned}\int_1^4 (t^{1/2} + t^{-1/2}) dt &= \left[\frac{2}{3}t^{3/2} + 2t^{1/2} \right]_1^4 \\ &= \left(\frac{2}{3}(4)^{3/2} + 2(4)^{1/2} \right) - \left(\frac{2}{3}(1)^{3/2} + 2(1)^{1/2} \right) \\ &= \left(\frac{16}{3} + 4 \right) - \left(\frac{2}{3} + 2 \right) = \left(\frac{28}{3} \right) - \left(\frac{8}{3} \right) = \frac{20}{3}\end{aligned}$$

Question 6

Evaluate the integral $\int_0^9 2\sqrt{t} dt$.

Solution Write the integrand as $2t^{1/2}$. An antiderivative is $F(t) = \frac{4}{3}t^{3/2}$.

$$\begin{aligned}\int_0^9 2\sqrt{t} dt &= \left[\frac{4}{3}t^{3/2} \right]_0^9 \\ &= \frac{4}{3}(9)^{3/2} - 0 = \frac{4}{3}(27) = \mathbf{36}\end{aligned}$$

Question 7

Evaluate the integral $\int_0^2 (u^5 - u^3 + u^2) du$.

Solution An antiderivative is $F(u) = \frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{3}u^3$.

$$\begin{aligned}\int_0^2 (u^5 - u^3 + u^2) du &= \left[\frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{3}u^3 \right]_0^2 \\ &= \left(\frac{64}{6} - \frac{16}{4} + \frac{8}{3} \right) - (0) = \frac{32}{3} - 4 + \frac{8}{3} \\ &= \frac{40}{3} - \frac{12}{3} = \mathbf{\frac{28}{3}}\end{aligned}$$

Question 8

Evaluate the integral $\int_1^2 \frac{y+5y^7}{y^3} dy$.

Solution Simplify the integrand: $\frac{y+5y^7}{y^3} = y^{-2} + 5y^4$. An antiderivative is $F(y) = -y^{-1} + y^5$.

$$\begin{aligned}\int_1^2 (y^{-2} + 5y^4) dy &= [-y^{-1} + y^5]_1^2 \\&= \left(-\frac{1}{2} + 2^5\right) - \left(-\frac{1}{1} + 1^5\right) \\&= \left(32 - \frac{1}{2}\right) - (-1 + 1) = 31.5 - 0 = \frac{63}{2}\end{aligned}$$

Question 9

Evaluate the integral $\int_1^9 \frac{3x-2}{\sqrt{x}} dx$.

Solution Simplify the integrand: $\frac{3x-2}{\sqrt{x}} = 3x^{1/2} - 2x^{-1/2}$. An antiderivative is $F(x) = 2x^{3/2} - 4x^{1/2}$.

$$\begin{aligned}\int_1^9 (3x^{1/2} - 2x^{-1/2}) dx &= [2x^{3/2} - 4x^{1/2}]_1^9 \\&= \left(2(9)^{3/2} - 4(9)^{1/2}\right) - \left(2(1)^{3/2} - 4(1)^{1/2}\right) \\&= (2(27) - 4(3)) - (2 - 4) \\&= (54 - 12) - (-2) = 42 + 2 = \mathbf{44}\end{aligned}$$

Question 10

Evaluate the integral $\int_0^\pi (4 \sin \theta - 3 \cos \theta) d\theta$.

Solution An antiderivative is $F(\theta) = -4 \cos \theta - 3 \sin \theta$.

$$\begin{aligned}\int_0^\pi (4 \sin \theta - 3 \cos \theta) d\theta &= [-4 \cos \theta - 3 \sin \theta]_0^\pi \\&= (-4 \cos \pi - 3 \sin \pi) - (-4 \cos 0 - 3 \sin 0) \\&= (-4(-1) - 0) - (-4(1) - 0) = 4 - (-4) = \mathbf{8}\end{aligned}$$

Question 11

Evaluate the integral $\int_{-1}^2 |x| dx$.

Solution We split the integral at $x = 0$:

$$\begin{aligned}\int_{-1}^2 |x| dx &= \int_{-1}^0 (-x) dx + \int_0^2 x dx \\&= \left[-\frac{x^2}{2}\right]_{-1}^0 + \left[\frac{x^2}{2}\right]_0^2 \\&= \left(0 - \left(-\frac{(-1)^2}{2}\right)\right) + \left(\frac{2^2}{2} - 0\right) \\&= \frac{1}{2} + 2 = \frac{5}{2}\end{aligned}$$

Question 12

Evaluate the integral $\int_{-\pi/2}^{\pi/2} \sin x dx$.

Solution The function $f(x) = \sin x$ is an odd function, and the interval $[-\pi/2, \pi/2]$ is symmetric around zero. Thus, the integral is zero.

$$\begin{aligned}\int_{-\pi/2}^{\pi/2} \sin x dx &= [-\cos x]_{-\pi/2}^{\pi/2} \\ &= (-\cos(\pi/2)) - (-\cos(-\pi/2)) = 0 - 0 = \mathbf{0}\end{aligned}$$

Question 13

Evaluate the integral $\int_0^3 \frac{1}{x-1} dx$.

Solution The function $f(x) = \frac{1}{x-1}$ has a discontinuity at $x = 1$, which is inside the interval $[4]$. This is an improper integral of Type 2. We split the integral: $\int_0^3 \frac{1}{x-1} dx = \int_0^1 \frac{1}{x-1} dx + \int_1^3 \frac{1}{x-1} dx$. The first integral is:

$$\int_0^1 \frac{1}{x-1} dx = \lim_{t \rightarrow 1^-} [\ln |x-1|]_0^t = \lim_{t \rightarrow 1^-} (\ln |t-1| - \ln(1))$$

As $t \rightarrow 1^-$, $\ln |t-1| \rightarrow -\infty$. Since the integral $\int_0^1 \frac{1}{x-1} dx$ **diverges**, the entire integral $\int_0^3 \frac{1}{x-1} dx$ **does not exist**.

Question 14

Evaluate the integral $\int_0^4 \frac{x}{\sqrt{25-x^2}} dx$.

Solution Let $u = 25 - x^2$. Then $du = -2x dx$, so $x dx = -\frac{1}{2} du$. Change limits: If $x = 0, u = 25 - 0^2 = 25$. If $x = 4, u = 25 - 4^2 = 9$.

$$\begin{aligned}\int_0^4 \frac{x}{\sqrt{25-x^2}} dx &= \int_{25}^9 \frac{1}{\sqrt{u}} \left(-\frac{1}{2}\right) du \\ &= -\frac{1}{2} \int_{25}^9 u^{-1/2} du = \frac{1}{2} \int_9^{25} u^{-1/2} du \\ &= \frac{1}{2} \left[2u^{1/2} \right]_9^{25} = [\sqrt{u}]_9^{25} = \sqrt{25} - \sqrt{9} = 5 - 3 = \mathbf{2}\end{aligned}$$

The integral is **convergent**.

Question 15

Evaluate the integral $\int_0^2 (x^3 - 3x + 3) dx$.

Solution An antiderivative is $F(x) = \frac{x^4}{4} - \frac{3x^2}{2} + 3x$.

$$\begin{aligned}\int_0^2 (x^3 - 3x + 3) dx &= \left[\frac{x^4}{4} - \frac{3x^2}{2} + 3x \right]_0^2 \\ &= \left(\frac{2^4}{4} - \frac{3(2^2)}{2} + 3(2) \right) - (0) \\ &= (4 - 6 + 6) = \mathbf{4}\end{aligned}$$

Question 16

Evaluate the integral $\int_0^{\pi/4} \sec^2 t dt$.

Solution An antiderivative of $f(t) = \sec^2 t$ is $F(t) = \tan t$.

$$\begin{aligned}\int_0^{\pi/4} \sec^2 t dt &= [\tan t]_0^{\pi/4} \\ &= \tan(\pi/4) - \tan(0) = 1 - 0 = \mathbf{1}\end{aligned}$$

The Fundamental Theorem of Calculus

Question 1

Find the derivative of the function $g(x) = \int_1^x t^2 dt$.

Solution We use the Fundamental Theorem of Calculus, Part 1 (FTC1), which states that if $g(x) = \int_a^x f(t)dt$, then $g'(x) = f(x)$. Here, $f(t) = t^2$ and $a = 1$.

$$g'(x) = \frac{d}{dx} \left(\int_1^x t^2 dt \right) = \mathbf{x^2}$$

Question 2

Find the derivative of the function $g(x) = \int_0^x (1 + \sqrt{t})dt$.

Solution Using FTC1 directly with $f(t) = 1 + \sqrt{t}$ and $a = 0$:

$$g'(x) = \frac{d}{dx} \left(\int_0^x (1 + \sqrt{t})dt \right) = \mathbf{1 + \sqrt{x}}$$

Question 3

Find the derivative of the function $g(x) = \int_0^x \sqrt{1 + 2t}dt$.

Solution Using FTC1 directly with $f(t) = \sqrt{1 + 2t}$:

$$g'(x) = \frac{d}{dx} \left(\int_0^x \sqrt{1 + 2t}dt \right) = \mathbf{\sqrt{1 + 2x}}$$

Question 4

Find the derivative of the function $g(x) = \int_1^x \ln t dt$.

Solution Using FTC1 directly with $f(t) = \ln t$:

$$g'(x) = \frac{d}{dx} \left(\int_1^x \ln t dt \right) = \mathbf{\ln x}$$

Question 5

Find the derivative of the function $g(y) = \int_2^y t^2 \sin t dt$.

Solution Using FTC1, noting that the independent variable is y :

$$g'(y) = \frac{d}{dy} \left(\int_2^y t^2 \sin t dt \right) = \mathbf{y^2 \sin y}$$

Question 6

Find the derivative of the function $g(u) = \int_3^u \frac{1}{x+x^2} dx$.

Solution Using FTC1, noting that the independent variable is u :

$$g'(u) = \frac{d}{du} \left(\int_3^u \frac{1}{x+x^2} dx \right) = \mathbf{\frac{1}{u+u^2}}$$

Question 7

Find the derivative of the function $y = \int_{1/x}^2 \frac{\cos t}{t} dt$.

Solution First, rewrite the integral using the property $\int_a^b f(t)dt = -\int_b^a f(t)dt$:

$$y = -\int_2^{1/x} \frac{\cos t}{t} dt$$

Now we use FTC1 combined with the Chain Rule. Let $u = 1/x$. Then $\frac{du}{dx} = -\frac{1}{x^2}$.

$$\begin{aligned}\frac{dy}{dx} &= -\frac{d}{du} \left(\int_2^u \frac{\cos t}{t} dt \right) \cdot \frac{du}{dx} \\ \frac{dy}{dx} &= -\left(\frac{\cos u}{u} \right) \cdot \left(-\frac{1}{x^2} \right)\end{aligned}$$

Substituting $u = 1/x$ back in:

$$\frac{dy}{dx} = -\left(\frac{\cos(1/x)}{1/x} \right) \left(-\frac{1}{x^2} \right) = \frac{\mathbf{x} \cos(\mathbf{1/x})}{\mathbf{x}^2} = \frac{\cos(\mathbf{1/x})}{\mathbf{x}}$$

Question 8

Find the derivative of the function $y = \int_1^{e^x} \frac{\ln t}{t} dt$.

Solution We use FTC1 combined with the Chain Rule. Let $u = e^x$. Then $\frac{du}{dx} = e^x$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{du} \left(\int_1^u \frac{\ln t}{t} dt \right) \cdot \frac{du}{dx} \\ \frac{dy}{dx} &= \left(\frac{\ln u}{u} \right) \cdot e^x\end{aligned}$$

Substituting $u = e^x$ back in, and noting $\ln(e^x) = x$:

$$\frac{dy}{dx} = \frac{\ln(e^x)}{e^x} \cdot e^x = \frac{x}{e^x} \cdot e^x = \mathbf{x}$$

Question 9

Evaluate the integral $\int_{-2}^3 6dx$.

Solution We use FTC2. An antiderivative of $f(x) = 6$ is $F(x) = 6x$.

$$\int_{-2}^3 6dx = [6x]_{-2}^3 = 6(3) - 6(-2) = 18 - (-12) = \mathbf{42}$$

Question 10

Evaluate the integral $\int_{-1}^2 (4x + 3)dx$.

Solution An antiderivative of $f(x) = 4x + 3$ is $F(x) = 2x^2 + 3x$.

$$\begin{aligned}\int_{-1}^2 (4x + 3)dx &= [2x^2 + 3x]_{-1}^2 \\ &= (2(2)^2 + 3(2)) - (2(-1)^2 + 3(-1)) \\ &= (8 + 6) - (2 - 3) = 14 - (-1) = \mathbf{15}\end{aligned}$$

Question 11

Evaluate the integral $\int_0^4 (1 + 3y - y^2)dy$.

Solution An antiderivative of $f(y) = 1 + 3y - y^2$ is $F(y) = y + \frac{3}{2}y^2 - \frac{1}{3}y^3$.

$$\begin{aligned}\int_0^4 (1 + 3y - y^2)dy &= \left[y + \frac{3}{2}y^2 - \frac{1}{3}y^3 \right]_0^4 \\ &= \left(4 + \frac{3}{2}(16) - \frac{1}{3}(64) \right) - (0) \\ &= 4 + 24 - \frac{64}{3} = 28 - \frac{64}{3} = \frac{84 - 64}{3} = \mathbf{\frac{20}{3}}\end{aligned}$$

Question 12

Evaluate the integral $\int_1^8 \frac{3}{\sqrt[3]{x}} dx$.

Solution First, rewrite the integrand as $3x^{-1/3}$. An antiderivative is $F(x) = 3 \cdot \frac{3}{2}x^{2/3} = \frac{9}{2}x^{2/3}$.

$$\begin{aligned}\int_1^8 3x^{-1/3} dx &= \left[\frac{9}{2}x^{2/3} \right]_1^8 \\ &= \frac{9}{2}(8)^{2/3} - \frac{9}{2}(1)^{2/3} \\ &= \frac{9}{2}(4) - \frac{9}{2}(1) = 18 - 4.5 = \frac{45}{2}\end{aligned}$$

Question 13

Evaluate the integral $\int_{-1}^1 x^5 dx$.

Solution An antiderivative of $f(x) = x^5$ is $F(x) = \frac{1}{6}x^6$.

$$\int_{-1}^1 x^5 dx = \left[\frac{1}{6}x^6 \right]_{-1}^1 = \frac{1}{6}(1)^6 - \frac{1}{6}(-1)^6 = \frac{1}{6} - \frac{1}{6} = 0$$

Note: Since $f(x) = x^5$ is an odd function and the interval of integration is symmetric around zero, the net area must be zero.

Question 14

Evaluate the integral $\int_0^2 (2x + x^6) dx$.

Solution An antiderivative of $f(x) = 2x + x^6$ is $F(x) = x^2 + \frac{1}{7}x^7$.

$$\begin{aligned}\int_0^2 (2x + x^6) dx &= \left[x^2 + \frac{1}{7}x^7 \right]_0^2 \\ &= \left(2^2 + \frac{1}{7}(2^7) \right) - (0) = 4 + \frac{128}{7} = \frac{28 + 128}{7} = \frac{156}{7}\end{aligned}$$

Question 15

Evaluate the integral $\int_1^4 \frac{1}{\sqrt{x}} dx$.

Solution Rewrite the integrand as $x^{-1/2}$. An antiderivative is $F(x) = 2x^{1/2}$.

$$\int_1^4 x^{-1/2} dx = [2\sqrt{x}]_1^4 = 2\sqrt{4} - 2\sqrt{1} = 2(2) - 2(1) = 4 - 2 = 2$$

Question 16

Evaluate the integral $\int_0^{\pi/4} \sec^2 t dt$.

Solution An antiderivative of $f(t) = \sec^2 t$ is $F(t) = \tan t$.

$$\int_0^{\pi/4} \sec^2 t dt = [\tan t]_0^{\pi/4} = \tan(\pi/4) - \tan(0) = 1 - 0 = 1$$

Question 17

Evaluate the integral $\int_0^2 (u^5 - u^3 + u^2) du$.

Solution

An antiderivative is $F(u) = \frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{3}u^3$.

$$\begin{aligned}\int_0^2 (u^5 - u^3 + u^2) du &= \left[\frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{3}u^3 \right]_0^2 \\ &= \left(\frac{64}{6} - \frac{16}{4} + \frac{8}{3} \right) - (0) = \frac{32}{3} - 4 + \frac{8}{3} = \frac{40}{3} - \frac{12}{3} = \frac{28}{3}\end{aligned}$$

Question 18

Evaluate the integral $\int_1^2 \frac{y+5y^7}{y^3} dy$.

Solution Simplify the integrand: $\frac{y+5y^7}{y^3} = y^{-2} + 5y^4$. An antiderivative is $F(y) = -y^{-1} + y^5$.

$$\begin{aligned}\int_1^2 (y^{-2} + 5y^4) dy &= [-y^{-1} + y^5]_1^2 \\ &= \left(-\frac{1}{2} + 32 \right) - (-1 + 1) = 31.5 - 0 = \frac{63}{2}\end{aligned}$$

Question 19

Evaluate the integral $\int_1^4 (t\sqrt{t} + \frac{1}{t\sqrt{t}}) dt$.

Solution Simplify the integrand: $t^{3/2} + t^{-3/2}$. An antiderivative is $F(t) = \frac{2}{5}t^{5/2} - 2t^{-1/2}$.

$$\begin{aligned}\int_1^4 (t^{3/2} + t^{-3/2}) dt &= \left[\frac{2}{5}t^{5/2} - \frac{2}{\sqrt{t}} \right]_1^4 \\ &= \left(\frac{2}{5}(32) - \frac{2}{2} \right) - \left(\frac{2}{5}(1) - \frac{2}{1} \right) \\ &= \left(\frac{64}{5} - 1 \right) - \left(\frac{2}{5} - 2 \right) = \frac{59}{5} - \left(-\frac{8}{5} \right) = \frac{67}{5} = 13.4\end{aligned}$$

Question 20

Evaluate the integral $\int_0^2 (x^2 + x - 1) dx$.

Solution An antiderivative is $F(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 - x$.

$$\begin{aligned}\int_0^2 (x^2 + x - 1) dx &= \left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - x \right]_0^2 \\ &= \left(\frac{8}{3} + \frac{4}{2} - 2 \right) - (0) = \frac{8}{3} + 2 - 2 = \frac{8}{3}\end{aligned}$$

Question 21

Evaluate the integral $\int_{-1}^2 (x^3 - 2x^2 + x - 1) dx$.

Solution An antiderivative is $F(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 - x$.

$$\begin{aligned}\int_{-1}^2 (x^3 - 2x^2 + x - 1) dx &= \left[\frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 - x \right]_{-1}^2 \\ &= \left(\frac{16}{4} - \frac{16}{3} + \frac{4}{2} - 2 \right) - \left(\frac{1}{4} - \frac{2}{3}(-1) + \frac{1}{2}(1) - (-1) \right)\end{aligned}$$

$$\begin{aligned}
&= \left(4 - \frac{16}{3} + 2 - 2\right) - \left(\frac{1}{4} + \frac{2}{3} + \frac{1}{2} + 1\right) \\
&= \left(4 - \frac{16}{3}\right) - \left(\frac{3 + 8 + 6 + 12}{12}\right) = \frac{12 - 16}{3} - \frac{29}{12} \\
&= -\frac{4}{3} - \frac{29}{12} = -\frac{16}{12} - \frac{29}{12} = -\frac{45}{12} = -\frac{15}{4}
\end{aligned}$$

Question 22

Evaluate the integral $\int_1^4 \frac{x^2+x+1}{\sqrt{x}} dx$.

Solution Simplify the integrand: $x^{3/2} + x^{1/2} + x^{-1/2}$. An antiderivative is $F(x) = \frac{2}{5}x^{5/2} + \frac{2}{3}x^{3/2} + 2x^{1/2}$.

$$\begin{aligned}
\int_1^4 (x^{3/2} + x^{1/2} + x^{-1/2}) dx &= \left[\frac{2}{5}x^{5/2} + \frac{2}{3}x^{3/2} + 2\sqrt{x} \right]_1^4 \\
&= \left(\frac{2}{5}(32) + \frac{2}{3}(8) + 2(2) \right) - \left(\frac{2}{5} + \frac{2}{3} + 2 \right) \\
&= \left(\frac{64}{5} + \frac{16}{3} + 4 \right) - \left(\frac{6 + 10 + 30}{15} \right) \\
&= \left(\frac{192 + 80 + 60}{15} \right) - \frac{46}{15} = \frac{332 - 46}{15} = \frac{286}{15}
\end{aligned}$$

Question 23

Evaluate the integral $\int_1^9 \frac{3x-2}{\sqrt{x}} dx$.

Solution Simplify the integrand: $3x^{1/2} - 2x^{-1/2}$. An antiderivative is $F(x) = 2x^{3/2} - 4x^{1/2}$.

$$\begin{aligned}
\int_1^9 (3x^{1/2} - 2x^{-1/2}) dx &= \left[2x^{3/2} - 4x^{1/2} \right]_1^9 \\
&= (2(27) - 4(3)) - (2(1) - 4(1)) = (54 - 12) - (-2) = 42 + 2 = 44
\end{aligned}$$

Question 24

Evaluate the integral $\int_0^\pi (4 \sin \theta - 3 \cos \theta) d\theta$.

Solution An antiderivative is $F(\theta) = -4 \cos \theta - 3 \sin \theta$.

$$\begin{aligned}
\int_0^\pi (4 \sin \theta - 3 \cos \theta) d\theta &= [-4 \cos \theta - 3 \sin \theta]_0^\pi \\
&= (-4 \cos \pi - 3 \sin \pi) - (-4 \cos 0 - 3 \sin 0) \\
&= (-4(-1) - 0) - (-4(1) - 0) = 4 - (-4) = 8
\end{aligned}$$

Question 25

Evaluate the integral $\int_1^9 \frac{1}{2x} dx$.

Solution

$$\begin{aligned}
\int_1^9 \frac{1}{2x} dx &= \frac{1}{2} \int_1^9 \frac{1}{x} dx = \frac{1}{2} [\ln |x|]_1^9 \\
&= \frac{1}{2} (\ln 9 - \ln 1) = \frac{1}{2} \ln 9
\end{aligned}$$

Question 26

Evaluate the integral $\int_0^{\pi/3} \sec t \tan t dt$.

Solution An antiderivative is $F(t) = \sec t$.

$$\begin{aligned}\int_0^{\pi/3} \sec t \tan t dt &= [\sec t]_0^{\pi/3} = \sec(\pi/3) - \sec(0) \\ &= 2 - 1 = \mathbf{1}\end{aligned}$$

Question 27

Evaluate the integral $\int_0^{\pi/4} \frac{1+\cos^2 x}{\cos^2 x} dx$.

Solution Split the integrand: $\frac{1+\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \sec^2 x + 1$.

$$\begin{aligned}\int_0^{\pi/4} (\sec^2 x + 1) dx &= [\tan x + x]_0^{\pi/4} \\ &= \left(\tan(\pi/4) + \frac{\pi}{4}\right) - (0) = \mathbf{1 + \frac{\pi}{4}}\end{aligned}$$

Question 28

Evaluate the integral $\int_1^2 \frac{1}{x^3} dx$.

Solution Rewrite the integrand as x^{-3} . An antiderivative is $F(x) = -\frac{1}{2}x^{-2}$.

$$\begin{aligned}\int_1^2 x^{-3} dx &= \left[-\frac{1}{2x^2}\right]_1^2 = \left(-\frac{1}{2(4)}\right) - \left(-\frac{1}{2(1)}\right) \\ &= -\frac{1}{8} + \frac{1}{2} = -\frac{1}{8} + \frac{4}{8} = \mathbf{\frac{3}{8}}\end{aligned}$$

Question 29

Evaluate the integral $\int_{-1}^2 |x| dx$. (Interpret the integral in terms of area.)

Solution The graph of $f(x) = |x|$ consists of two straight line segments forming a "V" shape. We split the integral at $x = 0$:

$$\int_{-1}^2 |x| dx = \int_{-1}^0 (-x) dx + \int_0^2 x dx$$

The first integral is a triangle with base 1 (from -1 to 0) and height 1 ($|-1| = 1$). Area $A_1 = \frac{1}{2} \times 1 \times 1 = 1/2$. The second integral is a triangle with base 2 (from 0 to 2) and height 2 ($|2| = 2$). Area $A_2 = \frac{1}{2} \times 2 \times 2 = 2$. Total Area = $A_1 + A_2 = \frac{1}{2} + 2 = \mathbf{\frac{5}{2}}$

Question 30

Explain why $\int_1^2 \sqrt{5-x} dx$ exists. (This involves confirming continuity, though a full calculation is not requested based on surrounding context. We proceed with the general interpretation or evaluation.)

Solution The definite integral $\int_a^b f(x) dx$ exists if $f(x)$ is continuous on the interval $[a, b]$. The function $f(x) = \sqrt{5-x}$ requires $5-x \geq 0$, so $x \leq 5$. Since the interval of integration is $[4, 5]$, and all values in this interval satisfy $x \leq 5$, the function $f(x)$ is **continuous** on the interval $[4, 5]$. Therefore, the integral **exists** (is convergent).

If requested to evaluate (using FTC2):

$$\begin{aligned}\int_1^2 \sqrt{5-x} dx &= \left[-\frac{2}{3}(5-x)^{3/2}\right]_1^2 \\ &= -\frac{2}{3}(5-2)^{3/2} - \left(-\frac{2}{3}(5-1)^{3/2}\right) \\ &= -\frac{2}{3}(3\sqrt{3}) + \frac{2}{3}(8) = \frac{16}{3} - 2\sqrt{3} = \mathbf{\frac{16-6\sqrt{3}}{3}}\end{aligned}$$

Question 31

Evaluate the integral $\int_0^\pi \cos x dx$ by interpreting it in terms of areas.

Solution The graph of $f(x) = \cos x$ on $[0, \pi]$ is above the x -axis on $[0, \pi/2]$ and below the x -axis on $[\pi/2, \pi]$. The definite integral represents the net area (A_{net}).

$$A_{net} = \text{Area above } x\text{-axis} - \text{Area below } x\text{-axis}$$

$$A_{net} = \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} \cos x dx$$

$$A_{net} = [\sin x]_0^{\pi/2} + [\sin x]_{\pi/2}^{\pi}$$

$$A_{net} = (1 - 0) + (0 - 1) = 1 - 1 = 0$$

(If the question intended to ask for the total area, $A_{total} = |\int_0^{\pi/2} \cos x dx| + |\int_{\pi/2}^{\pi} \cos x dx| = 1 + 1 = 2$. However, interpreting $\int \cos x dx$ as "net area" is the standard practice, yielding 0, which is also confirmed as a plausible calculation based on source geometry/area interpretations [6].)

Indefinite Integrals

Question 1

Evaluate the indefinite integral $\int (4x^3 - 2x^2 + 5) dx$.

Solution We use the constant multiple rule and the power rule for integration:

$$\begin{aligned} \int (4x^3 - 2x^2 + 5) dx &= 4 \int x^3 dx - 2 \int x^2 dx + 5 \int 1 dx \\ &= 4 \left(\frac{x^4}{4} \right) - 2 \left(\frac{x^3}{3} \right) + 5x + C = x^4 - \frac{2}{3}x^3 + 5x + C \end{aligned}$$

Question 2

Evaluate the indefinite integral $\int (x^5 + 6x^2 - 4) dx$.

Solution Applying the power rule:

$$\int (x^5 + 6x^2 - 4) dx = \frac{1}{6}x^6 + 2x^3 - 4x + C$$

Question 3

Evaluate the indefinite integral $\int (3x - 2)x^{-4} dx$.

Solution First, distribute and simplify the integrand: $(3x - 2)x^{-4} = 3x^{-3} - 2x^{-4}$. Applying the power rule:

$$\begin{aligned} \int (3x^{-3} - 2x^{-4}) dx &= 3 \left(\frac{x^{-2}}{-2} \right) - 2 \left(\frac{x^{-3}}{-3} \right) + C \\ &= -\frac{3}{2}x^{-2} + \frac{2}{3}x^{-3} + C = -\frac{3}{2x^2} + \frac{2}{3x^3} + C \end{aligned}$$

Question 4

Evaluate the indefinite integral $\int (3\sqrt{x} + 2/\sqrt{x}) dx$.

Solution Rewrite using rational exponents: $\int (3x^{1/2} + 2x^{-1/2}) dx$. Applying the power rule:

$$\begin{aligned} 3 \left(\frac{x^{3/2}}{3/2} \right) + 2 \left(\frac{x^{1/2}}{1/2} \right) + C &= 2x^{3/2} + 4x^{1/2} + C \\ &= 2x\sqrt{x} + 4\sqrt{x} + C \end{aligned}$$

Question 5

Evaluate the indefinite integral $\int (t^2 + 1/t^3)dt$.

Solution Rewrite as $t^2 + t^{-3}$ and apply the power rule:

$$\int (t^2 + t^{-3})dt = \frac{t^3}{3} + \frac{t^{-2}}{-2} + C = \frac{1}{3}t^3 - \frac{1}{2t^2} + C$$

Question 6

Evaluate the indefinite integral $\int (x + 1/x)dx$.

Solution We apply the power rule for the first term and the logarithmic rule for the second term:

$$\int (x + \frac{1}{x})dx = \frac{1}{2}x^2 + \ln |x| + C$$

Question 7

Evaluate the indefinite integral $\int \sqrt{x}(x-1)dx$.

Solution Distribute and rewrite using exponents: $\int (x^{3/2} - x^{1/2})dx$. Applying the power rule:

$$\frac{x^{5/2}}{5/2} - \frac{x^{3/2}}{3/2} + C = \frac{2}{5}x^{5/2} - \frac{2}{3}x^{3/2} + C$$

Question 8

Evaluate the indefinite integral $\int (e^x - 2x^2)dx$.

Solution Using the rule for exponential functions and the power rule:

$$\int (e^x - 2x^2)dx = e^x - \frac{2}{3}x^3 + C$$

Question 9

Evaluate the indefinite integral $\int (4 \sin x + 2 \cos x)dx$.

Solution Using the rules for trigonometric functions: $\int \sin x dx = -\cos x$ and $\int \cos x dx = \sin x$.

$$\int (4 \sin x + 2 \cos x)dx = -4 \cos x + 2 \sin x + C$$

Question 10

Evaluate the indefinite integral $\int (5/x - 3e^x)dx$.

Solution

$$5 \int \frac{1}{x}dx - 3 \int e^x dx = 5 \ln |x| - 3e^x + C$$

Definite Integrals

Question 1

Evaluate the integral $\int_{-2}^5 6dx$.

Solution Using FTC2:

$$\int_{-2}^5 6dx = [6x]_{-2}^5 = 6(5) - 6(-2) = 30 - (-12) = 42$$

Question 2

Evaluate the integral $\int_{-1}^2 (3 + 2x - 4x^3)dx$.

Solution An antiderivative is $F(x) = 3x + x^2 - x^4$.

$$\begin{aligned}\int_{-1}^2 (3 + 2x - 4x^3)dx &= [3x + x^2 - x^4]_{-1}^2 \\&= (3(2) + 2^2 - 2^4) - (3(-1) + (-1)^2 - (-1)^4) \\&= (6 + 4 - 16) - (-3 + 1 - 1) = -6 - (-3) = \mathbf{-3}\end{aligned}$$

Question 3

Evaluate the integral $\int_{-1}^2 (u^5 - u^3 + u^2)du$.

Solution An antiderivative is $F(u) = \frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{3}u^3$.

$$\begin{aligned}\int_{-1}^2 (u^5 - u^3 + u^2)du &= \left[\frac{1}{6}u^6 - \frac{1}{4}u^4 + \frac{1}{3}u^3 \right]_{-1}^2 \\&= \left(\frac{64}{6} - \frac{16}{4} + \frac{8}{3} \right) - \left(\frac{1}{6} - \frac{1}{4} - \frac{1}{3} \right) \\&= \left(\frac{32}{3} - 4 + \frac{8}{3} \right) - \left(\frac{2 - 3 - 4}{12} \right) = \left(\frac{40}{3} - \frac{12}{3} \right) - \left(-\frac{5}{12} \right) \\&= \frac{28}{3} + \frac{5}{12} = \frac{112 + 5}{12} = \frac{\mathbf{117}}{\mathbf{12}} = \frac{\mathbf{39}}{\mathbf{4}}\end{aligned}$$

Question 4

Evaluate the integral $\int_{-2}^2 (3u + 1)^2 du$.

Solution Expand the integrand: $(3u + 1)^2 = 9u^2 + 6u + 1$. An antiderivative is $F(u) = 3u^3 + 3u^2 + u$.

$$\begin{aligned}\int_{-2}^2 (9u^2 + 6u + 1)du &= [3u^3 + 3u^2 + u]_{-2}^2 \\&= (3(8) + 3(4) + 2) - (3(-8) + 3(4) - 2) \\&= (24 + 12 + 2) - (-24 + 12 - 2) = 38 - (-14) = \mathbf{52}\end{aligned}$$

Question 5

Evaluate the integral $\int_0^4 (2v + 5)(3v - 1)dv$.

Solution Expand the integrand: $(2v + 5)(3v - 1) = 6v^2 + 13v - 5$. An antiderivative is $F(v) = 2v^3 + \frac{13}{2}v^2 - 5v$.

$$\begin{aligned}\int_0^4 (6v^2 + 13v - 5)dv &= \left[2v^3 + \frac{13}{2}v^2 - 5v \right]_0^4 \\&= \left(2(64) + \frac{13}{2}(16) - 5(4) \right) - (0) \\&= 128 + 104 - 20 = \mathbf{212}\end{aligned}$$

Question 6

Evaluate the integral $\int_1^4 \sqrt{t}(1+t)dt$.

Solution Simplify the integrand: $t^{1/2} + t^{3/2}$. An antiderivative is $F(t) = \frac{2}{3}t^{3/2} + \frac{2}{5}t^{5/2}$.

$$\begin{aligned}\int_1^4 (t^{1/2} + t^{3/2})dt &= \left[\frac{2}{3}t^{3/2} + \frac{2}{5}t^{5/2} \right]_1^4 \\&= \left(\frac{2}{3}(8) + \frac{2}{5}(32) \right) - \left(\frac{2}{3}(1) + \frac{2}{5}(1) \right) \\&= \frac{16}{3} + \frac{64}{5} - \frac{2}{3} - \frac{2}{5} = \left(\frac{16-2}{3} \right) + \left(\frac{64-2}{5} \right) = \frac{14}{3} + \frac{62}{5} \\&= \frac{70+186}{15} = \frac{\mathbf{256}}{\mathbf{15}}\end{aligned}$$

Question 7

Evaluate the integral $\int_0^9 2\sqrt{t}dt$.

Solution Rewrite as $2t^{1/2}$. An antiderivative is $F(t) = 2 \cdot \frac{2}{3}t^{3/2} = \frac{4}{3}t^{3/2}$.

$$\int_0^9 2\sqrt{t}dt = \left[\frac{4}{3}t^{3/2} \right]_0^9 = \frac{4}{3}(9^{3/2}) - 0 = \frac{4}{3}(27) = \mathbf{36}$$

Question 8

Evaluate the integral $\int_1^2 \frac{y+5y^7}{y^3}dy$.

Solution Simplify the integrand: $\frac{y}{y^3} + \frac{5y^7}{y^3} = y^{-2} + 5y^4$. An antiderivative is $F(y) = -y^{-1} + y^5$.

$$\begin{aligned}\int_1^2 (y^{-2} + 5y^4)dy &= [-y^{-1} + y^5]_1^2 \\&= \left(-\frac{1}{2} + 32 \right) - (-1 + 1) = 31.5 - 0 = \frac{\mathbf{63}}{\mathbf{2}}\end{aligned}$$

Question 9

Evaluate the integral $\int_0^1 x(x^{3/2} + x^{5/4})dx$.

Solution Simplify the integrand: $x^{5/2} + x^{9/4}$. An antiderivative is $F(x) = \frac{2}{7}x^{7/2} + \frac{4}{13}x^{13/4}$.

$$\begin{aligned}\int_0^1 (x^{5/2} + x^{9/4})dx &= \left[\frac{2}{7}x^{7/2} + \frac{4}{13}x^{13/4} \right]_0^1 \\&= \left(\frac{2}{7} + \frac{4}{13} \right) - (0) = \frac{26+28}{91} = \frac{\mathbf{54}}{\mathbf{91}}\end{aligned}$$

Question 10

Evaluate the integral $\int_0^5 (2e^x + 4\cos x)dx$.

Solution An antiderivative is $F(x) = 2e^x + 4\sin x$.

$$\begin{aligned}\int_0^5 (2e^x + 4\cos x)dx &= [2e^x + 4\sin x]_0^5 \\&= (2e^5 + 4\sin 5) - (2e^0 + 4\sin 0) = \mathbf{2e^5 + 4\sin 5 - 2}\end{aligned}$$

[5]

Question 11

Evaluate the integral $\int_1^9 \frac{3x-2}{\sqrt{x}} dx$.

Solution Simplify the integrand: $3x^{1/2} - 2x^{-1/2}$. An antiderivative is $F(x) = 2x^{3/2} - 4x^{1/2}$.

$$\begin{aligned}\int_1^9 (3x^{1/2} - 2x^{-1/2}) dx &= \left[2x^{3/2} - 4x^{1/2} \right]_1^9 \\ &= (2(27) - 4(3)) - (2(1) - 4(1)) = 42 - (-2) = \mathbf{44}\end{aligned}$$

Question 12

Evaluate the integral $\int_0^\pi (4 \sin \theta - 3 \cos \theta) d\theta$.

Solution An antiderivative is $F(\theta) = -4 \cos \theta - 3 \sin \theta$.

$$\begin{aligned}\int_0^\pi (4 \sin \theta - 3 \cos \theta) d\theta &= [-4 \cos \theta - 3 \sin \theta]_0^\pi \\ &= (-4(-1) - 0) - (-4(1) - 0) = 4 - (-4) = \mathbf{8}\end{aligned}$$

Question 13

Evaluate the integral $\int_{\pi/4}^{\pi/3} \sec t \tan t dt$.

Solution An antiderivative is $F(t) = \sec t$.

$$\begin{aligned}\int_{\pi/4}^{\pi/3} \sec t \tan t dt &= [\sec t]_{\pi/4}^{\pi/3} = \sec(\pi/3) - \sec(\pi/4) \\ &= 2 - \sqrt{2}\end{aligned}$$

Question 14

Evaluate the integral $\int_0^{\pi/4} \frac{1+\cos^2 x}{\cos^2 x} dx$.

Solution Simplify the integrand: $\sec^2 x + 1$. An antiderivative is $F(x) = \tan x + x$.

$$\begin{aligned}\int_0^{\pi/4} (\sec^2 x + 1) dx &= [\tan x + x]_0^{\pi/4} \\ &= \left(\tan(\pi/4) + \frac{\pi}{4} \right) - (0) = \mathbf{1 + \frac{\pi}{4}}\end{aligned}$$

Question 15

Evaluate the integral $\int_1^9 \frac{1}{2x} dx$.

Solution

$$\begin{aligned}\int_1^9 \frac{1}{2x} dx &= \frac{1}{2} \int_1^9 \frac{1}{x} dx = \frac{1}{2} [\ln |x|]_1^9 \\ &= \frac{1}{2} (\ln 9 - \ln 1) = \frac{\mathbf{1}}{\mathbf{2}} \ln \mathbf{9}\end{aligned}$$

Question 16

Evaluate the integral $\int_0^{\pi/3} \sec t \tan t dt$.

Solution This is identical to Question 29, but with different limits.

$$\int_0^{\pi/3} \sec t \tan t dt = [\sec t]_0^{\pi/3} = \sec(\pi/3) - \sec(0) = 2 - 1 = \mathbf{1}$$

Question 17

Evaluate the integral $\int_{-1}^1 x^5 dx$.

Solution The integrand $f(x) = x^5$ is an odd function, and the interval is symmetric around zero.

$$\int_{-1}^1 x^5 dx = \left[\frac{1}{6} x^6 \right]_{-1}^1 = \frac{1}{6}(1)^6 - \frac{1}{6}(-1)^6 = \frac{1}{6} - \frac{1}{6} = \mathbf{0}$$

Question 18

Evaluate the integral $\int_{-1}^2 |x| dx$.

Solution We split the integral at $x = 0$ where the definition of $|x|$ changes:

$$\begin{aligned} \int_{-1}^2 |x| dx &= \int_{-1}^0 (-x) dx + \int_0^2 x dx \\ &= \left[-\frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_0^2 \\ &= \left(0 - \left(-\frac{1}{2} \right) \right) + \left(\frac{4}{2} - 0 \right) = \frac{1}{2} + 2 = \mathbf{\frac{5}{2}} \end{aligned}$$

Question 19

Evaluate the integral $\int_0^4 \frac{x}{\sqrt{25-x^2}} dx$.

Solution This requires u-substitution. Let $u = 25 - x^2$, so $du = -2x dx$ or $x dx = -\frac{1}{2} du$. Limits: $x = 0 \implies u = 25$; $x = 4 \implies u = 9$.

$$\begin{aligned} \int_{25}^9 \frac{1}{\sqrt{u}} \left(-\frac{1}{2} du \right) &= \frac{1}{2} \int_9^{25} u^{-1/2} du \\ &= \frac{1}{2} [2u^{1/2}]_9^{25} = [\sqrt{u}]_9^{25} = 5 - 3 = \mathbf{2} \end{aligned}$$

Question 20

Evaluate the integral $\int_0^2 (x^3 - 3x + 3) dx$.

Solution An antiderivative is $F(x) = \frac{1}{4}x^4 - \frac{3}{2}x^2 + 3x$.

$$\begin{aligned} \int_0^2 (x^3 - 3x + 3) dx &= \left[\frac{1}{4}x^4 - \frac{3}{2}x^2 + 3x \right]_0^2 \\ &= \left(\frac{16}{4} - \frac{3(4)}{2} + 3(2) \right) - (0) = 4 - 6 + 6 = \mathbf{4} \end{aligned}$$

Question 21

Evaluate the integral $\int_{\pi/4}^{\pi/3} \sec^2 t dt$.

Solution An antiderivative is $F(t) = \tan t$.

$$\int_{\pi/4}^{\pi/3} \sec^2 t dt = [\tan t]_{\pi/4}^{\pi/3} = \tan(\pi/3) - \tan(\pi/4) = \sqrt{3} - \mathbf{1}$$

Question 22

Evaluate the integral $\int_0^1 (1+x^2)^3 dx$.

Solution Expand the integrand: $(1+x^2)^3 = 1 + 3x^2 + 3x^4 + x^6$. An antiderivative is $F(x) = x + x^3 + \frac{3}{5}x^5 + \frac{1}{7}x^7$.

$$\begin{aligned}\int_0^1 (1+3x^2+3x^4+x^6)dx &= \left[x + x^3 + \frac{3}{5}x^5 + \frac{1}{7}x^7 \right]_0^1 \\ &= \left(1 + 1 + \frac{3}{5} + \frac{1}{7} \right) - (0) = 2 + \frac{21+5}{35} = 2 + \frac{26}{35} \\ &= \frac{70+26}{35} = \frac{96}{35}\end{aligned}$$

Question 23

Evaluate the integral $\int_0^{\pi/2} (2\cos x - 5x)dx$.

Solution An antiderivative is $F(x) = 2\sin x - \frac{5}{2}x^2$.

$$\begin{aligned}\int_0^{\pi/2} (2\cos x - 5x)dx &= \left[2\sin x - \frac{5}{2}x^2 \right]_0^{\pi/2} \\ &= \left(2\sin(\pi/2) - \frac{5}{2}\left(\frac{\pi}{2}\right)^2 \right) - (0) \\ &= 2(1) - \frac{5}{2}\left(\frac{\pi^2}{4}\right) = 2 - \frac{5\pi^2}{8}\end{aligned}$$

Method of Substitution

Question 1

Evaluate $\int x^2\sqrt{x^3+1}dx$.

Solution Let $u = x^3 + 1$. Then the differential is $du = 3x^2dx$. Solving for x^2dx , we get $x^2dx = \frac{1}{3}du$. Substitute u and du into the integral:

$$\int u^{1/2} \left(\frac{1}{3}du \right) = \frac{1}{3} \int u^{1/2} du$$

Apply the Power Rule:

$$= \frac{1}{3} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} + C$$

Substitute back $u = x^3 + 1$:

$$= \frac{2}{9} (x^3 + 1)^{3/2} + C$$

Question 2

Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

Solution Let $u = \sqrt{x}$. Then the differential is $du = \frac{1}{2\sqrt{x}}dx$. Solving for $dx/\sqrt{x}$, we get $2du = \frac{1}{\sqrt{x}}dx$. Substitute u and du into the integral:

$$\int \sin u (2du) = 2 \int \sin u du$$

Apply the trigonometric integral rule:

$$= 2(-\cos u) + C$$

Substitute back $u = \sqrt{x}$:

$$= -2\cos \sqrt{x} + C$$

Question 3

Evaluate $\int \frac{4}{(1+2x)^3} dx$.

Solution Rewrite the integrand: $4 \int (1+2x)^{-3} dx$. Let $u = 1+2x$. Then $du = 2dx$, or $dx = \frac{1}{2}du$. Substitute u and du into the integral:

$$4 \int u^{-3} \left(\frac{1}{2} du \right) = 2 \int u^{-3} du$$

Apply the Power Rule:

$$= 2 \cdot \frac{u^{-2}}{-2} + C = -u^{-2} + C$$

Substitute back $u = 1+2x$:

$$= -\frac{1}{(1+2x)^2} + C$$

Question 4

Evaluate $\int (3x-2)^{20} dx$.

Solution Let $u = 3x-2$. Then $du = 3dx$, or $dx = \frac{1}{3}du$. Substitute u and du into the integral:

$$\int u^{20} \left(\frac{1}{3} du \right) = \frac{1}{3} \int u^{20} du$$

Apply the Power Rule:

$$= \frac{1}{3} \cdot \frac{u^{21}}{21} + C = \frac{1}{63} u^{21} + C$$

Substitute back $u = 3x-2$:

$$= \frac{1}{63} (3x-2)^{21} + C$$

Question 5

Evaluate $\int \frac{1}{x \ln x} dx$.

Solution Let $u = \ln x$. Then the differential is $du = \frac{1}{x} dx$. Substitute u and du into the integral:

$$\int \frac{1}{u} du$$

Apply the logarithmic integral rule:

$$= \ln |u| + C$$

Substitute back $u = \ln x$:

$$= \ln |\ln x| + C$$

Question 6

Evaluate $\int \frac{x}{x^2+1} dx$.

Solution Let $u = x^2+1$. Then $du = 2x dx$, or $x dx = \frac{1}{2} du$. Substitute u and du into the integral:

$$\int \frac{1}{u} \left(\frac{1}{2} du \right) = \frac{1}{2} \int \frac{1}{u} du$$

Apply the logarithmic integral rule:

$$= \frac{1}{2} \ln |u| + C$$

Substitute back $u = x^2+1$. Since x^2+1 is always positive, the absolute value is optional:

$$= \frac{1}{2} \ln(x^2+1) + C$$

Question 7

Evaluate $\int \frac{3}{(2y+1)^5} dy$.

Solution Rewrite the integrand: $3 \int (2y+1)^{-5} dy$. Let $u = 2y + 1$. Then $du = 2dy$, or $dy = \frac{1}{2}du$. Substitute u and du into the integral:

$$3 \int u^{-5} \left(\frac{1}{2} du \right) = \frac{3}{2} \int u^{-5} du$$

Apply the Power Rule:

$$= \frac{3}{2} \cdot \frac{u^{-4}}{-4} + C = -\frac{3}{8} u^{-4} + C$$

Substitute back $u = 2y + 1$:

$$= -\frac{3}{8(2y+1)^4} + C$$

Question 8

Evaluate $\int e^{\cos t} \sin t dt$.

Solution Let $u = \cos t$. Then the differential is $du = -\sin t dt$, or $\sin t dt = -du$. Substitute u and du into the integral:

$$\int e^u (-du) = - \int e^u du$$

Apply the exponential integral rule:

$$= -e^u + C$$

Substitute back $u = \cos t$:

$$= -e^{\cos t} + C$$

Question 9

Evaluate $\int \sqrt{4-t} dt$.

Solution Let $u = 4 - t$. Then $du = -dt$, or $dt = -du$. Substitute u and du into the integral:

$$\int u^{1/2} (-du) = - \int u^{1/2} du$$

Apply the Power Rule:

$$= -\frac{u^{3/2}}{3/2} + C = -\frac{2}{3} u^{3/2} + C$$

Substitute back $u = 4 - t$:

$$= -\frac{2}{3} (4-t)^{3/2} + C$$

Question 10

Evaluate $\int y^3 \sqrt{2y^4+1} dy$.

Solution Let $u = 2y^4 + 1$. Then $du = 8y^3 dy$, or $y^3 dy = \frac{1}{8} du$. Substitute u and du into the integral:

$$\int u^{1/2} \left(\frac{1}{8} du \right) = \frac{1}{8} \int u^{1/2} du$$

Apply the Power Rule:

$$= \frac{1}{8} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{12} u^{3/2} + C$$

Substitute back $u = 2y^4 + 1$:

$$= \frac{1}{12} (2y^4+1)^{3/2} + C$$

Question 11

Evaluate $\int_0^{1/2} \frac{\arcsin x}{\sqrt{1-x^2}} dx$.

Solution Let $u = \arcsin x$. Then $du = \frac{1}{\sqrt{1-x^2}} dx$. Change the limits: When $x = 0$, $u = \arcsin(0) = 0$. When $x = 1/2$, $u = \arcsin(1/2) = \pi/6$. Substitute u and du and the new limits into the integral:

$$\int_0^{\pi/6} u du$$

Apply the Power Rule:

$$\begin{aligned} &= \left[\frac{u^2}{2} \right]_0^{\pi/6} \\ &= \frac{1}{2} \left(\left(\frac{\pi}{6} \right)^2 - 0^2 \right) = \frac{1}{2} \cdot \frac{\pi^2}{36} = \frac{\pi^2}{72} \end{aligned}$$

Question 12

Evaluate $\int_0^1 x^2(1+2x^3)^5 dx$.

Solution Let $u = 1 + 2x^3$. Then $du = 6x^2 dx$, or $x^2 dx = \frac{1}{6} du$. Change the limits: When $x = 0$, $u = 1 + 2(0)^3 = 1$. When $x = 1$, $u = 1 + 2(1)^3 = 3$. Substitute u and du and the new limits into the integral:

$$\int_1^3 u^5 \left(\frac{1}{6} du \right) = \frac{1}{6} \int_1^3 u^5 du$$

Apply the Power Rule:

$$\begin{aligned} &= \frac{1}{6} \left[\frac{u^6}{6} \right]_1^3 = \frac{1}{36} [3^6 - 1^6] \\ &= \frac{1}{36} (729 - 1) = \frac{728}{36} = \frac{182}{9} \end{aligned}$$

Question 13

Evaluate $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$.

Solution Let $u = \ln x$. Then $du = \frac{1}{x} dx$. Change the limits: When $x = e$, $u = \ln e = 1$. When $x = e^4$, $u = \ln e^4 = 4$. Substitute u and du and the new limits into the integral:

$$\int_1^4 u^{-1/2} du$$

Apply the Power Rule:

$$\begin{aligned} &= \left[\frac{u^{1/2}}{1/2} \right]_1^4 = [2\sqrt{u}]_1^4 \\ &= 2\sqrt{4} - 2\sqrt{1} = 2(2) - 2(1) = 2 \end{aligned}$$

Question 14

Evaluate $\int_0^{\pi/2} \cos x \sin(\sin x) dx$.

Solution Let $u = \sin x$. Then $du = \cos x dx$. Change the limits: When $x = 0$, $u = \sin(0) = 0$. When $x = \pi/2$, $u = \sin(\pi/2) = 1$. Substitute u and du and the new limits into the integral:

$$\int_0^1 \sin u du$$

Apply the trigonometric integral rule:

$$\begin{aligned} &= [-\cos u]_0^1 \\ &= -\cos(1) - (-\cos(0)) = 1 - \cos(1) \end{aligned}$$

(The term $\cos(1)$ is measured in radians.)

$$= 1 - \cos(1)$$

Question 15

Evaluate $\int_0^a x\sqrt{a^2 - x^2}dx$, where $a > 0$.

Solution Let $u = a^2 - x^2$. Then $du = -2xdx$, or $xdx = -\frac{1}{2}du$. Change the limits: When $x = 0$, $u = a^2 - 0^2 = a^2$. When $x = a$, $u = a^2 - a^2 = 0$. Substitute u and du and the new limits into the integral:

$$\int_{a^2}^0 u^{1/2} \left(-\frac{1}{2}du\right) = \frac{1}{2} \int_0^{a^2} u^{1/2} du$$

(We reversed the limits and absorbed the negative sign.) Apply the Power Rule:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{u^{3/2}}{3/2} \right]_0^{a^2} = \frac{1}{2} \cdot \frac{2}{3} [u^{3/2}]_0^{a^2} \\ &= \frac{1}{3} [(a^2)^{3/2} - 0] = \frac{1}{3} \mathbf{a^3} \end{aligned}$$

Question 16

Evaluate $\int_0^{\pi/4} \sec^2 t \tan^2 t dt$.

Solution Let $u = \tan t$. Then $du = \sec^2 t dt$. Change the limits: When $t = 0$, $u = \tan(0) = 0$. When $t = \pi/4$, $u = \tan(\pi/4) = 1$. Substitute u and du and the new limits into the integral:

$$\int_0^1 u^2 du$$

Apply the Power Rule:

$$\begin{aligned} &= \left[\frac{u^3}{3} \right]_0^1 \\ &= \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3} \end{aligned}$$

Question 17

Evaluate $\int x^2 \sqrt{x^3 + 1} dx$.

Solution Let $u = x^3 + 1$. Then the differential is $du = 3x^2 dx$. Solving for $x^2 dx$, we get $x^2 dx = \frac{1}{3} du$. Substitute u and du into the integral:

$$\int u^{1/2} \left(\frac{1}{3} du\right) = \frac{1}{3} \int u^{1/2} du$$

Apply the Power Rule:

$$= \frac{1}{3} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} + C$$

Substitute back $u = x^3 + 1$:

$$= \frac{2}{9} (\mathbf{x^3 + 1})^{3/2} + \mathbf{C}$$

Question 18

Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

Solution Let $u = \sqrt{x}$. Then the differential is $du = \frac{1}{2\sqrt{x}} dx$. Solving for $dx/\sqrt{x}$, we get $2du = \frac{1}{\sqrt{x}} dx$. Substitute u and du into the integral:

$$\int \sin u (2du) = 2 \int \sin u du$$

Apply the trigonometric integral rule:

$$= 2(-\cos u) + C$$

Substitute back $u = \sqrt{x}$:

$$= -\mathbf{2 \cos \sqrt{x} + C}$$

Question 19

Evaluate $\int \frac{4}{(1+2x)^3} dx$.

Solution Rewrite the integrand: $4 \int (1+2x)^{-3} dx$. Let $u = 1+2x$. Then $du = 2dx$, or $dx = \frac{1}{2}du$. Substitute u and du into the integral:

$$4 \int u^{-3} \left(\frac{1}{2} du \right) = 2 \int u^{-3} du$$

Apply the Power Rule:

$$= 2 \cdot \frac{u^{-2}}{-2} + C = -u^{-2} + C$$

Substitute back $u = 1+2x$:

$$= -\frac{1}{(1+2x)^2} + C$$

Question 20

Evaluate $\int (3x-2)^{20} dx$.

Solution Let $u = 3x-2$. Then $du = 3dx$, or $dx = \frac{1}{3}du$. Substitute u and du into the integral:

$$\int u^{20} \left(\frac{1}{3} du \right) = \frac{1}{3} \int u^{20} du$$

Apply the Power Rule:

$$= \frac{1}{3} \cdot \frac{u^{21}}{21} + C = \frac{1}{63} u^{21} + C$$

Substitute back $u = 3x-2$:

$$= \frac{1}{63} (3x-2)^{21} + C$$

Question 21

Evaluate $\int \frac{1}{x \ln x} dx$.

Solution Let $u = \ln x$. Then the differential is $du = \frac{1}{x} dx$. Substitute u and du into the integral:

$$\int \frac{1}{u} du$$

Apply the logarithmic integral rule:

$$= \ln |u| + C$$

Substitute back $u = \ln x$:

$$= \ln |\ln x| + C$$

Question 22

Evaluate $\int \frac{x}{x^2+1} dx$.

Solution Let $u = x^2+1$. Then $du = 2xdx$, or $xdx = \frac{1}{2}du$. Substitute u and du into the integral:

$$\int \frac{1}{u} \left(\frac{1}{2} du \right) = \frac{1}{2} \int \frac{1}{u} du$$

Apply the logarithmic integral rule:

$$= \frac{1}{2} \ln |u| + C$$

Substitute back $u = x^2+1$. Since x^2+1 is always positive, the absolute value is optional:

$$= \frac{1}{2} \ln(x^2+1) + C$$

Question 23

Evaluate $\int \frac{3}{(2y+1)^5} dy$.

Solution Rewrite the integrand: $3 \int (2y+1)^{-5} dy$. Let $u = 2y + 1$. Then $du = 2dy$, or $dy = \frac{1}{2}du$. Substitute u and du into the integral:

$$3 \int u^{-5} \left(\frac{1}{2} du \right) = \frac{3}{2} \int u^{-5} du$$

Apply the Power Rule:

$$= \frac{3}{2} \cdot \frac{u^{-4}}{-4} + C = -\frac{3}{8} u^{-4} + C$$

Substitute back $u = 2y + 1$:

$$= -\frac{3}{8(2y+1)^4} + C$$

Question 24

Evaluate $\int e^{\cos t} \sin t dt$.

Solution Let $u = \cos t$. Then the differential is $du = -\sin t dt$, or $\sin t dt = -du$. Substitute u and du into the integral:

$$\int e^u (-du) = - \int e^u du$$

Apply the exponential integral rule:

$$= -e^u + C$$

Substitute back $u = \cos t$:

$$= -e^{\cos t} + C$$

Question 25

Evaluate $\int \sqrt{4-t} dt$.

Solution Let $u = 4 - t$. Then $du = -dt$, or $dt = -du$. Substitute u and du into the integral:

$$\int u^{1/2} (-du) = - \int u^{1/2} du$$

Apply the Power Rule:

$$= -\frac{u^{3/2}}{3/2} + C = -\frac{2}{3} u^{3/2} + C$$

Substitute back $u = 4 - t$:

$$= -\frac{2}{3} (4-t)^{3/2} + C$$

Question 26

Evaluate $\int y^3 \sqrt{2y^4+1} dy$.

Solution Let $u = 2y^4 + 1$. Then $du = 8y^3 dy$, or $y^3 dy = \frac{1}{8} du$. Substitute u and du into the integral:

$$\int u^{1/2} \left(\frac{1}{8} du \right) = \frac{1}{8} \int u^{1/2} du$$

Apply the Power Rule:

$$= \frac{1}{8} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{12} u^{3/2} + C$$

Substitute back $u = 2y^4 + 1$:

$$= \frac{1}{12} (2y^4+1)^{3/2} + C$$

Question 27

Evaluate $\int_0^{1/2} \frac{\arcsin x}{\sqrt{1-x^2}} dx$.

Solution Let $u = \arcsin x$. Then $du = \frac{1}{\sqrt{1-x^2}} dx$. Change the limits: When $x = 0$, $u = \arcsin(0) = 0$. When $x = 1/2$, $u = \arcsin(1/2) = \pi/6$. Substitute u and du and the new limits into the integral:

$$\int_0^{\pi/6} u du$$

Apply the Power Rule:

$$\begin{aligned} &= \left[\frac{u^2}{2} \right]_0^{\pi/6} \\ &= \frac{1}{2} \left(\left(\frac{\pi}{6} \right)^2 - 0^2 \right) = \frac{1}{2} \cdot \frac{\pi^2}{36} = \frac{\pi^2}{72} \end{aligned}$$

Question 28

Evaluate $\int_0^1 x^2(1+2x^3)^5 dx$.

Solution Let $u = 1 + 2x^3$. Then $du = 6x^2 dx$, or $x^2 dx = \frac{1}{6} du$. Change the limits: When $x = 0$, $u = 1 + 2(0)^3 = 1$. When $x = 1$, $u = 1 + 2(1)^3 = 3$. Substitute u and du and the new limits into the integral:

$$\int_1^3 u^5 \left(\frac{1}{6} du \right) = \frac{1}{6} \int_1^3 u^5 du$$

Apply the Power Rule:

$$\begin{aligned} &= \frac{1}{6} \left[\frac{u^6}{6} \right]_1^3 = \frac{1}{36} [3^6 - 1^6] \\ &= \frac{1}{36} (729 - 1) = \frac{728}{36} = \frac{182}{9} \end{aligned}$$

Question 29

Evaluate $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$.

Solution Let $u = \ln x$. Then $du = \frac{1}{x} dx$. Change the limits: When $x = e$, $u = \ln e = 1$. When $x = e^4$, $u = \ln e^4 = 4$. Substitute u and du and the new limits into the integral:

$$\int_1^4 u^{-1/2} du$$

Apply the Power Rule:

$$\begin{aligned} &= \left[\frac{u^{1/2}}{1/2} \right]_1^4 = [2\sqrt{u}]_1^4 \\ &= 2\sqrt{4} - 2\sqrt{1} = 2(2) - 2(1) = \mathbf{2} \end{aligned}$$

Question 30

Evaluate $\int_0^{\pi/2} \cos x \sin(\sin x) dx$.

Solution Let $u = \sin x$. Then $du = \cos x dx$. Change the limits: When $x = 0$, $u = \sin(0) = 0$. When $x = \pi/2$, $u = \sin(\pi/2) = 1$. Substitute u and du and the new limits into the integral:

$$\int_0^1 \sin u du$$

Apply the trigonometric integral rule:

$$\begin{aligned} &= [-\cos u]_0^1 \\ &= -\cos(1) - (-\cos(0)) = 1 - \cos(1) \end{aligned}$$

(The term $\cos(1)$ is measured in radians.)

$$= \mathbf{1 - \cos(1)}$$

Question 31

Evaluate $\int_0^a x\sqrt{a^2 - x^2}dx$, where $a > 0$.

Solution Let $u = a^2 - x^2$. Then $du = -2xdx$, or $xdx = -\frac{1}{2}du$. Change the limits: When $x = 0$, $u = a^2 - 0^2 = a^2$. When $x = a$, $u = a^2 - a^2 = 0$. Substitute u and du and the new limits into the integral:

$$\int_{a^2}^0 u^{1/2} \left(-\frac{1}{2}du\right) = \frac{1}{2} \int_0^{a^2} u^{1/2} du$$

(We reversed the limits and absorbed the negative sign.) Apply the Power Rule:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{u^{3/2}}{3/2} \right]_0^{a^2} = \frac{1}{2} \cdot \frac{2}{3} [u^{3/2}]_0^{a^2} \\ &= \frac{1}{3} [(a^2)^{3/2} - 0] = \frac{1}{3} \mathbf{a^3} \end{aligned}$$

Question 32

Evaluate $\int_0^{\pi/4} \sec^2 t \tan^2 t dt$.

Solution Let $u = \tan t$. Then $du = \sec^2 t dt$. Change the limits: When $t = 0$, $u = \tan(0) = 0$. When $t = \pi/4$, $u = \tan(\pi/4) = 1$. Substitute u and du and the new limits into the integral:

$$\int_0^1 u^2 du$$

Apply the Power Rule:

$$\begin{aligned} &= \left[\frac{u^3}{3} \right]_0^1 \\ &= \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3} \end{aligned}$$

Trigonometric Integrals

Question 1

Evaluate the indefinite integral $\int \sin^3(mx)dx$.

Solution Since the power of sine is odd (3), we save one $\sin(mx)$ factor and express the remaining factors in terms of cosine using the identity $\sin^2 \theta = 1 - \cos^2 \theta$.

$$\int \sin^3(mx)dx = \int \sin^2(mx) \sin(mx)dx = \int (1 - \cos^2(mx)) \sin(mx)dx$$

Let $u = \cos(mx)$. Then $du = -m \sin(mx)dx$, which means $\sin(mx)dx = -\frac{1}{m}du$. Substituting u and du :

$$\begin{aligned} &= \int (1 - u^2) \left(-\frac{1}{m}du\right) = -\frac{1}{m} \int (1 - u^2)du \\ &= -\frac{1}{m} \left(u - \frac{u^3}{3}\right) + C = -\frac{1}{m}u + \frac{1}{3m}u^3 + C \end{aligned}$$

Substituting back $u = \cos(mx)$:

$$= -\frac{1}{\mathbf{m}} \cos(\mathbf{mx}) + \frac{1}{\mathbf{3m}} \cos^3(\mathbf{mx}) + \mathbf{C}$$

Question 2

Evaluate the definite integral $\int_0^{\pi/2} \cos^2 \theta d\theta$.

Solution Since the power of cosine is even, we use the half-angle identity $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$ [2, 3].

$$\begin{aligned}\int_0^{\pi/2} \cos^2 \theta d\theta &= \int_0^{\pi/2} \frac{1}{2}(1 + \cos 2\theta) d\theta \\ &= \frac{1}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2}\end{aligned}$$

Applying the Fundamental Theorem of Calculus (FTC):

$$\begin{aligned}&= \frac{1}{2} \left[\left(\frac{\pi}{2} + \frac{1}{2} \sin(\pi) \right) - \left(0 + \frac{1}{2} \sin(0) \right) \right] \\ &= \frac{1}{2} \left[\left(\frac{\pi}{2} + 0 \right) - (0 + 0) \right] = \frac{\pi}{4}\end{aligned}$$

Question 3

Evaluate the definite integral $\int_0^{\pi/2} \sin^2(2\theta) d\theta$.

Solution Since the power of sine is even, we use the half-angle identity $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$. Here, $x = 2\theta$, so $2x = 4\theta$.

$$\begin{aligned}\int_0^{\pi/2} \sin^2(2\theta) d\theta &= \int_0^{\pi/2} \frac{1}{2}(1 - \cos 4\theta) d\theta \\ &= \frac{1}{2} \left[\theta - \frac{1}{4} \sin 4\theta \right]_0^{\pi/2}\end{aligned}$$

Applying the FTC:

$$\begin{aligned}&= \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{1}{4} \sin(2\pi) \right) - \left(0 - \frac{1}{4} \sin(0) \right) \right] \\ &= \frac{1}{2} \left[\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right] = \frac{\pi}{4}\end{aligned}$$

Question 4

Evaluate the definite integral $\int_0^{\pi/4} \sin^4 x \cos^2 x dx$.

Solution Since both powers are even, we use the half-angle identities $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, along with the identity $\sin x \cos x = \frac{1}{2} \sin 2x$.

$$\begin{aligned}\int_0^{\pi/4} (\sin^2 x \cos^2 x) \sin^2 x dx &= \int_0^{\pi/4} \left(\frac{1}{4} \sin^2 2x \right) \left(\frac{1}{2} (1 - \cos 2x) \right) dx \\ &= \frac{1}{8} \int_0^{\pi/4} (\sin^2 2x - \sin^2 2x \cos 2x) dx\end{aligned}$$

We split the integral:

$$\begin{aligned}I_1 &= \frac{1}{8} \int_0^{\pi/4} \sin^2 2x dx = \frac{1}{16} \int_0^{\pi/4} (1 - \cos 4x) dx \\ I_1 &= \frac{1}{16} \left[x - \frac{1}{4} \sin 4x \right]_0^{\pi/4} = \frac{1}{16} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{64} \\ I_2 &= \frac{1}{8} \int_0^{\pi/4} \sin^2 2x \cos 2x dx\end{aligned}$$

Let $u = \sin 2x$, $du = 2 \cos 2x dx$. Limits: $x = 0 \rightarrow u = 0$, $x = \pi/4 \rightarrow u = 1$.

$$I_2 = \frac{1}{8} \int_0^1 u^2 \left(\frac{1}{2} du \right) = \frac{1}{16} \left[\frac{u^3}{3} \right]_0^1 = \frac{1}{48}$$

$$\text{Total Integral} = I_1 - I_2 = \frac{\pi}{64} - \frac{1}{48} = \frac{3\pi - 4}{192}$$

Question 5

Evaluate the indefinite integral $\int \cos \theta \cos^5(\sin \theta) d\theta$.

Solution This integral involves a composite function. We start with the substitution $u = \sin \theta$.

$$\int \cos \theta \cos^5(\sin \theta) d\theta$$

Let $u = \sin \theta$. Then $du = \cos \theta d\theta$.

$$= \int \cos^5 u du$$

Since the power of cosine (5) is odd, we save one $\cos u$ factor and convert the remaining $\cos^4 u$ to sine using $\cos^2 u = 1 - \sin^2 u$:

$$= \int (\cos^2 u)^2 \cos u du = \int (1 - \sin^2 u)^2 \cos u du$$

Now let $w = \sin u$. Then $dw = \cos u du$.

$$= \int (1 - w^2)^2 dw = \int (1 - 2w^2 + w^4) dw$$

$$= w - \frac{2}{3}w^3 + \frac{1}{5}w^5 + C$$

Substituting back $w = \sin u$ and $u = \sin \theta$:

$$= \sin(\sin \theta) - \frac{2}{3} \sin^3(\sin \theta) + \frac{1}{5} \sin^5(\sin \theta) + \mathbf{C}$$

Question 6

Evaluate the indefinite integral $\int \tan^2 x dx$.

Solution This involves a power of tangent. We use the identity $\tan^2 x = \sec^2 x - 1$.

$$\int \tan^2 x dx = \int (\sec^2 x - 1) dx$$

We know that $\int \sec^2 x dx = \tan x + C$.

$$= \tan x - x + \mathbf{C}$$

Question 7

Evaluate $\int \frac{\sqrt{a^2 - x^2}}{x^2} dx$.

Solution We use the substitution for the form $\sqrt{a^2 - x^2}$: Let $x = a \sin \theta$. Then $dx = a \cos \theta d\theta$ and $\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = a \cos \theta$.

$$\begin{aligned} \int \frac{\sqrt{a^2 - x^2}}{x^2} dx &= \int \frac{a \cos \theta}{a^2 \sin^2 \theta} (a \cos \theta d\theta) \\ &= \int \frac{a^2 \cos^2 \theta}{a^2 \sin^2 \theta} d\theta = \int \cot^2 \theta d\theta \end{aligned}$$

Using the identity $\cot^2 \theta = \csc^2 \theta - 1$:

$$= \int (\csc^2 \theta - 1) d\theta = -\cot \theta - \theta + C$$

Since $x = a \sin \theta$, we have $\sin \theta = x/a$, so $\theta = \arcsin(x/a)$ and $\cot \theta = \frac{\sqrt{a^2 - x^2}}{x}$.

$$= -\frac{\sqrt{a^2 - x^2}}{x} - \arcsin\left(\frac{x}{a}\right) + \mathbf{C}$$

Question 8

Evaluate $\int \frac{x^3 dx}{\sqrt{x^2+9}}$. (Assuming the standard problem form for this index number in Section 7.3.)

Solution We use the substitution for the form $\sqrt{x^2+a^2}$ where $a = 3$: Let $\mathbf{x} = 3 \tan \theta$. Then $d\mathbf{x} = 3 \sec^2 \theta d\theta$ and $\sqrt{x^2+9} = \sqrt{9 \tan^2 \theta + 9} = 3 \sec \theta$.

$$\begin{aligned}\int \frac{x^3}{\sqrt{x^2+9}} dx &= \int \frac{(3 \tan \theta)^3}{3 \sec \theta} (3 \sec^2 \theta d\theta) \\ &= \int \frac{27 \tan^3 \theta}{3 \sec \theta} (3 \sec^2 \theta d\theta) = 27 \int \tan^3 \theta \sec \theta d\theta\end{aligned}$$

We save one $\sec \theta \tan \theta$ factor:

$$= 27 \int \tan^2 \theta (\sec \theta \tan \theta) d\theta = 27 \int (\sec^2 \theta - 1) (\sec \theta \tan \theta) d\theta$$

Let $u = \sec \theta$, $du = \sec \theta \tan \theta d\theta$:

$$= 27 \int (u^2 - 1) du = 27 \left(\frac{u^3}{3} - u \right) + C = 9u^3 - 27u + C$$

Substitute back $u = \sec \theta = \frac{\sqrt{x^2+9}}{3}$:

$$\begin{aligned}&= 9 \left(\frac{(x^2+9)^{3/2}}{27} \right) - 27 \left(\frac{\sqrt{x^2+9}}{3} \right) + C \\ &= \frac{1}{3} (x^2+9)^{3/2} - 9\sqrt{x^2+9} + C\end{aligned}$$

This can be factored:

$$= \frac{1}{3} \sqrt{\mathbf{x}^2+9} (\mathbf{x}^2 - 18) + \mathbf{C}$$

Question 9

Evaluate $\int \frac{dx}{x\sqrt{25-x^2}}$.

Solution We use the substitution for the form $\sqrt{a^2-x^2}$ where $a = 5$: Let $\mathbf{x} = 5 \sin \theta$. Then $d\mathbf{x} = 5 \cos \theta d\theta$ and $\sqrt{25-x^2} = 5 \cos \theta$.

$$\begin{aligned}\int \frac{dx}{x\sqrt{25-x^2}} &= \int \frac{5 \cos \theta d\theta}{(5 \sin \theta)(5 \cos \theta)} \\ &= \int \frac{1}{5 \sin \theta} d\theta = \frac{1}{5} \int \csc \theta d\theta\end{aligned}$$

Using the integral rule $\int \csc \theta d\theta = \ln |\csc \theta - \cot \theta|$:

$$= \frac{1}{5} \ln |\csc \theta - \cot \theta| + C$$

Since $x = 5 \sin \theta$, $\csc \theta = 5/x$ and $\cot \theta = \frac{\sqrt{25-x^2}}{x}$.

$$= \frac{1}{5} \ln \left| \frac{5 - \sqrt{25-x^2}}{x} \right| + \mathbf{C}$$

Question 10

Evaluate $\int \frac{\sqrt{x^2-9}}{x^3} dx$.

Solution We use the substitution for the form $\sqrt{x^2-a^2}$ where $a = 3$: Let $\mathbf{x} = 3 \sec \theta$. Then $d\mathbf{x} = 3 \sec \theta \tan \theta d\theta$ and $\sqrt{x^2-9} = 3 \tan \theta$.

$$\int \frac{\sqrt{x^2-9}}{x^3} dx = \int \frac{3 \tan \theta}{(3 \sec \theta)^3} (3 \sec \theta \tan \theta d\theta)$$

$$= \int \frac{9 \sec \theta \tan^2 \theta}{27 \sec^3 \theta} d\theta = \frac{1}{3} \int \frac{\tan^2 \theta}{\sec^2 \theta} d\theta$$

Since $\frac{\tan^2 \theta}{\sec^2 \theta} = \sin^2 \theta$, we use the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$:

$$\begin{aligned} &= \frac{1}{3} \int \sin^2 \theta d\theta = \frac{1}{6} \int (1 - \cos 2\theta) d\theta \\ &= \frac{1}{6} \left(\theta - \frac{1}{2} \sin 2\theta \right) + C = \frac{1}{6} (\theta - \sin \theta \cos \theta) + C \end{aligned}$$

Since $x = 3 \sec \theta$, $\theta = \operatorname{arcsec}(x/3)$, $\sin \theta = \frac{\sqrt{x^2-9}}{x}$ and $\cos \theta = 3/x$.

$$= \frac{1}{6} \left[\operatorname{arcsec} \left(\frac{x}{3} \right) - \frac{3\sqrt{x^2-9}}{x^2} \right] + C$$

Question 11

Evaluate $\int \frac{dt}{(t^2-6t+13)^{3/2}}$.

Solution First, complete the square in the denominator: $t^2 - 6t + 13 = (t^2 - 6t + 9) + 4 = (t - 3)^2 + 2^2$. Let $\mathbf{u} = \mathbf{t} - \mathbf{3}$. Then $du = dt$. The integral becomes:

$$\int \frac{du}{(u^2 + 2^2)^{3/2}}$$

Now use the trigonometric substitution for $\sqrt{u^2 + a^2}$, where $a = 2$: Let $\mathbf{u} = \mathbf{2} \tan \theta$. Then $d\mathbf{u} = \mathbf{2} \sec^2 \theta d\theta$ and $u^2 + 4 = 4 \tan^2 \theta + 4 = 4 \sec^2 \theta$.

$$\begin{aligned} \int \frac{2 \sec^2 \theta d\theta}{(4 \sec^2 \theta)^{3/2}} &= \int \frac{2 \sec^2 \theta d\theta}{8 \sec^3 \theta} = \frac{1}{4} \int \frac{1}{\sec \theta} d\theta \\ &= \frac{1}{4} \int \cos \theta d\theta = \frac{1}{4} \sin \theta + C \end{aligned}$$

Since $u = 2 \tan \theta$, we have $\tan \theta = u/2$. Constructing a right triangle gives $\sin \theta = \frac{u}{\sqrt{u^2+4}}$. Substitute back $u = t - 3$:

$$\begin{aligned} &= \frac{1}{4} \frac{u}{\sqrt{u^2+4}} + C = \frac{1}{4} \frac{t-3}{\sqrt{(t-3)^2+4}} + C \\ &= \frac{\mathbf{t} - \mathbf{3}}{4\sqrt{\mathbf{t}^2 - 6\mathbf{t} + 13}} + C \end{aligned}$$

Question 12

Evaluate $\int \frac{dx}{\sqrt{9x^2+6x-8}}$.

Solution First, complete the square under the radical:

$$9x^2 + 6x - 8 = (9x^2 + 6x + 1) - 9 = (3x + 1)^2 - 3^2$$

Let $\mathbf{u} = \mathbf{3x} + \mathbf{1}$. Then $du = 3dx$, or $d\mathbf{x} = \frac{1}{3}d\mathbf{u}$. The integral becomes:

$$\int \frac{1}{3} \frac{du}{\sqrt{u^2 - 3^2}}$$

Using the hyperbolic identity $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1}(u/a) + C$ or the logarithmic form $\ln |u + \sqrt{u^2 - a^2}| + C$ (Reference [1] uses the logarithmic form in its solution of a similar type integral $\int \frac{du}{\sqrt{u^2 - a^2}}$):

$$= \frac{1}{3} \ln |u + \sqrt{u^2 - 9}| + C$$

Substitute back $u = 3x + 1$:

$$= \frac{1}{3} \ln |3x + 1 + \sqrt{9x^2 + 6x - 8}| + C$$

Integration by Parts

Question 1

Evaluate $\int x \cos(5x) dx$.

Solution We select u and dv such that u simplifies upon differentiation and dv is easily integrable. Let $u = x$, so $du = dx$. Let $dv = \cos(5x)dx$, so $v = \frac{1}{5} \sin(5x)$. Applying the integration by parts formula:

$$\begin{aligned}\int x \cos(5x) dx &= x \left(\frac{1}{5} \sin(5x) \right) - \int \frac{1}{5} \sin(5x) dx \\ &= \frac{1}{5} x \sin(5x) - \frac{1}{5} \int \sin(5x) dx\end{aligned}$$

We integrate the remaining term: $\int \sin(5x) dx = -\frac{1}{5} \cos(5x)$.

$$\begin{aligned}&= \frac{1}{5} x \sin(5x) - \frac{1}{5} \left(-\frac{1}{5} \cos(5x) \right) + C \\ &= \frac{1}{5} \mathbf{x} \sin(\mathbf{5x}) + \frac{1}{25} \cos(\mathbf{5x}) + \mathbf{C}\end{aligned}$$

Question 2

Evaluate $\int x^2 e^x dx$.

Solution This requires repeated application of integration by parts (or using the tabular method/reduction formulas [1]). **First Application:** Let $u = x^2$, so $du = 2x dx$. Let $dv = e^x dx$, so $v = e^x$.

$$\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx = x^2 e^x - 2 \int x e^x dx$$

Second Application (on $\int x e^x dx$): Let $U = x$, so $dU = dx$. Let $dV = e^x dx$, so $V = e^x$.

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C_1$$

Substitute back into the main integral:

$$\begin{aligned}\int x^2 e^x dx &= x^2 e^x - 2(x e^x - e^x) + C \\ &= \mathbf{e^x(x^2 - 2x + 2)} + \mathbf{C}\end{aligned}$$

Question 3

Evaluate $\int \ln(2x+1) dx$.

Solution The logarithm function simplifies greatly when differentiated. Let $u = \ln(2x+1)$, so $du = \frac{2}{2x+1} dx$. Let $dv = dx$, so $v = x$.

$$\begin{aligned}\int \ln(2x+1) dx &= x \ln(2x+1) - \int x \left(\frac{2}{2x+1} \right) dx \\ &= x \ln(2x+1) - \int \frac{2x}{2x+1} dx\end{aligned}$$

Use algebraic manipulation on the remaining integral: $\frac{2x}{2x+1} = \frac{(2x+1)-1}{2x+1} = 1 - \frac{1}{2x+1}$.

$$\begin{aligned}&= x \ln(2x+1) - \int \left(1 - \frac{1}{2x+1} \right) dx \\ &= x \ln(2x+1) - \left(x - \frac{1}{2} \ln|2x+1| \right) + C \\ &= \mathbf{x \ln(2x+1) - x + \frac{1}{2} \ln(2x+1) + C}\end{aligned}$$

Question 4

Evaluate $\int \arcsin x dx$.

Solution Let $u = \arcsin x$, so $du = \frac{1}{\sqrt{1-x^2}}dx$. Let $dv = dx$, so $v = x$.

$$\int \arcsin x dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx$$

We evaluate the remaining integral using substitution: Let $w = 1 - x^2$, so $dw = -2x dx$, or $x dx = -\frac{1}{2}dw$.

$$\int \frac{x}{\sqrt{1-x^2}} dx = \int w^{-1/2} \left(-\frac{1}{2}dw\right) = -\frac{1}{2}(2w^{1/2}) + C_1 = -\sqrt{1-x^2} + C_1$$

Substitute back into the main integral:

$$\begin{aligned} \int \arcsin x dx &= x \arcsin x - (-\sqrt{1-x^2}) + C \\ &= x \arcsin x + \sqrt{1-x^2} + C \end{aligned}$$

Question 5

Evaluate $\int x \cos(x) dx$.

Solution Let $u = x$, so $du = dx$. Let $dv = \cos(x)dx$, so $v = \sin(x)$.

$$\begin{aligned} \int x \cos(x) dx &= x \sin(x) - \int \sin(x) dx \\ &= x \sin(x) + \cos(x) + C \end{aligned}$$

Question 6

Evaluate $\int x \sec^2(x) dx$.

Solution Let $u = x$, so $du = dx$. Let $dv = \sec^2(x)dx$, so $v = \tan(x)$.

$$\int x \sec^2(x) dx = x \tan(x) - \int \tan(x) dx$$

Using the known indefinite integral $\int \tan x dx = \ln |\sec x| + C$:

$$= x \tan(x) - \ln |\sec x| + C$$

Question 7

Evaluate $\int_0^1 x e^{-x} dx$.

Solution Let $u = x$, so $du = dx$. Let $dv = e^{-x}dx$, so $v = -e^{-x}$.

$$\int x e^{-x} dx = -x e^{-x} - \int (-e^{-x}) dx = -x e^{-x} - e^{-x} + C = -e^{-x}(x+1) + C$$

Applying FTC2:

$$\begin{aligned} \int_0^1 x e^{-x} dx &= [-e^{-x}(x+1)]_0^1 \\ &= (-e^{-1}(1+1)) - (-e^0(0+1)) \\ &= -2e^{-1} - (-1) = 1 - \frac{2}{e} \end{aligned}$$

Question 8

Evaluate $\int_1^4 \sqrt{x} \ln x dx$.

Solution Let $u = \ln x$, so $du = \frac{1}{x} dx$. Let $dv = x^{1/2} dx$, so $v = \frac{2}{3} x^{3/2}$.

$$\begin{aligned}\int \sqrt{x} \ln x dx &= \frac{2}{3} x^{3/2} \ln x - \int \frac{2}{3} x^{3/2} \left(\frac{1}{x} \right) dx \\&= \frac{2}{3} x^{3/2} \ln x - \frac{2}{3} \int x^{1/2} dx = \frac{2}{3} x^{3/2} \ln x - \frac{2}{3} \left(\frac{2}{3} x^{3/2} \right) + C \\&= \frac{2}{3} x^{3/2} \ln x - \frac{4}{9} x^{3/2} + C\end{aligned}$$

Applying FTC2:

$$\begin{aligned}\int_1^4 \sqrt{x} \ln x dx &= \left[\frac{2}{3} x^{3/2} \ln x - \frac{4}{9} x^{3/2} \right]_1^4 \\&= \left(\frac{2}{3} (4^{3/2}) \ln 4 - \frac{4}{9} (4^{3/2}) \right) - \left(\frac{2}{3} (1) \ln 1 - \frac{4}{9} (1) \right) \\&= \left(\frac{2}{3} (8) \ln 4 - \frac{4}{9} (8) \right) - \left(0 - \frac{4}{9} \right) \\&= \frac{16}{3} \ln 4 - \frac{32}{9} + \frac{4}{9} = \frac{16}{3} \ln 4 - \frac{28}{9}\end{aligned}$$

Question 9

Evaluate $\int_0^\pi x \sin(x) dx$.

Solution Let $u = x$, so $du = dx$. Let $dv = \sin(x) dx$, so $v = -\cos(x)$.

$$\int x \sin(x) dx = -x \cos(x) - \int (-\cos(x)) dx = -x \cos(x) + \sin(x) + C$$

Applying FTC2:

$$\begin{aligned}\int_0^\pi x \sin(x) dx &= [-x \cos(x) + \sin(x)]_0^\pi \\&= (-\pi \cos(\pi) + \sin(\pi)) - (0 \cos(0) + \sin(0)) \\&= (-\pi(-1) + 0) - (0) = \pi\end{aligned}$$

Question 10

Evaluate $\int_0^1 \arctan x dx$.

Solution Let $u = \arctan x$, so $du = \frac{1}{1+x^2} dx$. Let $dv = dx$, so $v = x$.

$$\int \arctan x dx = x \arctan x - \int \frac{x}{1+x^2} dx$$

We evaluate the remaining integral using substitution ($w = 1 + x^2$, $dw = 2x dx$):

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{1}{w} dw = \frac{1}{2} \ln(1+x^2) + C_1$$

Substitute back into the main integral:

$$\begin{aligned}\int_0^1 \arctan x dx &= \left[x \arctan x - \frac{1}{2} \ln(1+x^2) \right]_0^1 \\&= \left(1 \arctan(1) - \frac{1}{2} \ln(1+1^2) \right) - \left(0 \arctan(0) - \frac{1}{2} \ln(1) \right) \\&= \left(1 \cdot \frac{\pi}{4} - \frac{1}{2} \ln 2 \right) - (0 - 0) = \frac{\pi}{4} - \frac{1}{2} \ln 2\end{aligned}$$

Question 11

Evaluate $\int_0^t e^s \sin(t-s) ds$.

Solution We treat t as a constant. This integral requires two applications of parts, similar to $\int e^{ax} \sin(bx) dx$. Let $u = \sin(t-s)$, so $du = -\cos(t-s)ds$ (Chain Rule w.r.t s). Let $dv = e^s ds$, so $v = e^s$.

$$I = \int_0^t e^s \sin(t-s) ds = [e^s \sin(t-s)]_0^t - \int_0^t e^s (-\cos(t-s)) ds$$

$$I = (e^t \sin(0) - e^0 \sin(t)) + \int_0^t e^s \cos(t-s) ds$$

$$I = -\sin(t) + I_1$$

Now apply parts to $I_1 = \int_0^t e^s \cos(t-s) ds$: Let $U = \cos(t-s)$, so $dU = \sin(t-s)ds$. Let $dV = e^s ds$, so $V = e^s$.

$$I_1 = [e^s \cos(t-s)]_0^t - \int_0^t e^s \sin(t-s) ds$$

$$I_1 = (e^t \cos(0) - e^0 \cos(t)) - I$$

$$I_1 = e^t - \cos(t) - I$$

Substitute I_1 back into the expression for I :

$$I = -\sin(t) + (e^t - \cos(t) - I)$$

$$2I = e^t - \cos(t) - \sin(t)$$

$$\int_0^t e^s \sin(t-s) ds = \frac{1}{2}(e^t - \cos t - \sin t)$$

Question 12

Prove the reduction formula for $\int (\ln x)^n dx$.

Solution Let $I_n = \int (\ln x)^n dx$. We choose u and dv to reduce the power of the logarithm upon integration. Let $u = (\ln x)^n$, so $du = n(\ln x)^{n-1} \cdot \frac{1}{x} dx$. Let $dv = dx$, so $v = x$.

$$I_n = x(\ln x)^n - \int x \left(n(\ln x)^{n-1} \cdot \frac{1}{x} \right) dx$$

The x terms cancel:

$$I_n = x(\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$\int (\ln x)^n dx = x(\ln x)^n - n \int (\ln x)^{n-1} dx$$

Question 13

Prove the reduction formula for $\int x^n e^x dx$.

Solution Let $I_n = \int x^n e^x dx$. We choose u and dv to reduce the power of x upon differentiation. Let $u = x^n$, so $du = nx^{n-1} dx$. Let $dv = e^x dx$, so $v = e^x$.

$$I_n = x^n e^x - \int nx^{n-1} e^x dx$$

$$\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$$

Question 14

Prove the reduction formula for $\int \sec^n x dx$ (where $n \geq 2$).

Solution Let $I_n = \int \sec^n x dx$. We split the integral as $\int \sec^{n-2} x \cdot \sec^2 x dx$. Let $u = \sec^{n-2} x$, so $du = (n-2) \sec^{n-3} x \cdot (\sec x \tan x) dx$. Let $dv = \sec^2 x dx$, so $v = \tan x$.

$$I_n = \tan x \sec^{n-2} x - \int \tan x (n-2) \sec^{n-2} x \tan x dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x \tan^2 x dx$$

Using the identity $\tan^2 x = \sec^2 x - 1$:

$$I_n = \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \int (\sec^n x - \sec^{n-2} x) dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) I_n + (n-2) \int \sec^{n-2} x dx$$

Solving for I_n :

$$I_n + (n-2) I_n = \tan x \sec^{n-2} x + (n-2) \int \sec^{n-2} x dx$$

$$(1+n-2) I_n = \tan x \sec^{n-2} x + (n-2) \int \sec^{n-2} x dx$$

$$(n-1) I_n = \tan x \sec^{n-2} x + (n-2) \int \sec^{n-2} x dx$$

$$\int \sec^n x dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x dx$$

Section 6.1: Areas Between Curves

Question 15

Find the area of the region bounded by $y = x^2$ and $y = 2x$.

Solution The curves intersect when $x^2 = 2x$, giving $x(x-2) = 0$, so the intersection points are $x = 0$ and $x = 2$. On the interval $[1]$, the upper curve is $y = 2x$ and the lower curve is $y = x^2$.

$$A = \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2$$

$$A = \left(4 - \frac{8}{3} \right) - (0) = \frac{12-8}{3} = \frac{4}{3}$$

Question 16

Find the area of the region bounded by $y = x^2$ and $y = x^4$.

Solution The curves intersect when $x^4 = x^2$, giving $x^2(x^2 - 1) = 0$, so the intersection points are $x = -1$, $x = 0$, and $x = 1$. Since the region is symmetric about the y -axis and $x^2 \geq x^4$ on $[-1, 1]$:

$$A = \int_{-1}^1 (x^2 - x^4) dx = 2 \int_0^1 (x^2 - x^4) dx$$

$$A = 2 \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = 2 \left(\frac{1}{3} - \frac{1}{5} \right) = 2 \left(\frac{5-3}{15} \right) = \frac{4}{15}$$

Question 17

Find the area of the region bounded by $y = x^2$ and $y = \sqrt{x}$.

Solution The curves intersect when $x^2 = \sqrt{x}$, or $x^4 = x$, giving $x(x^3 - 1) = 0$. The intersection points are $x = 0$ and $x = 1$. On the interval $[2]$, the upper curve is $y = \sqrt{x}$ and the lower curve is $y = x^2$.

$$A = \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right]_0^1$$
$$A = \left(\frac{2}{3}(1) - \frac{1}{3}(1) \right) - (0) = \frac{1}{3}$$

Question 18

Find the area of the region bounded by $y = x^3$ and $y = x$.

Solution The curves intersect when $x^3 = x$, giving $x(x^2 - 1) = 0$, so $x = -1$, $x = 0$, and $x = 1$. Since $y = x$ is the upper function on $[2]$ and the region is symmetric (odd function boundaries):

$$A = 2 \int_0^1 (x - x^3) dx = 2 \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1$$
$$A = 2 \left(\frac{1}{2} - \frac{1}{4} \right) = 2 \left(\frac{1}{4} \right) = \frac{1}{2}$$

Question 19

Find the area of the region bounded by $y = 12 - x^2$ and $y = x^2 - 6$.

Solution The curves intersect when $12 - x^2 = x^2 - 6$, giving $2x^2 = 18$, or $x^2 = 9$. Intersections are at $x = -3$ and $x = 3$. Since $y = 12 - x^2$ is the upper function on $[-3, 3]$:

$$A = \int_{-3}^3 ((12 - x^2) - (x^2 - 6)) dx = \int_{-3}^3 (18 - 2x^2) dx$$
$$A = 2 \left[18x - \frac{2x^3}{3} \right]_0^3 = 2 \left(18(3) - \frac{2(27)}{3} \right)$$
$$A = 2(54 - 18) = 2(36) = \mathbf{72}$$

Question 20

Find the area of the region bounded by $y = x^3$ and $y = 3x - 2$.

Solution The curves intersect at $x^3 = 3x - 2$, which factors as $(x - 1)^2(x + 2) = 0$. The roots are $x = -2$ and $x = 1$. On the interval $[-2, 1]$, the upper curve is $y = x^3$.

$$A = \int_{-2}^1 (x^3 - (3x - 2)) dx = \left[\frac{x^4}{4} - \frac{3x^2}{2} + 2x \right]_{-2}^1$$
$$A = \left(\frac{1}{4} - \frac{3}{2} + 2 \right) - (4 - 6 - 4) = \frac{3}{4} - (-6) = \frac{27}{4}$$

Question 21

Find the area of the region bounded by $y = \sqrt{x}$, $y = 1/x$, and $x = 4$.

Solution The curves $y = \sqrt{x}$ and $y = 1/x$ intersect when $\sqrt{x} = 1/x$, so $x^{3/2} = 1$, or $x = 1$. The region extends from $x = 1$ to $x = 4$.

$$A = \int_1^4 (\sqrt{x} - \frac{1}{x}) dx = \left[\frac{2}{3} x^{3/2} - \ln|x| \right]_1^4$$
$$A = \left(\frac{2}{3}(8) - \ln 4 \right) - \left(\frac{2}{3} - \ln 1 \right) = \frac{16}{3} - \ln 4 - \frac{2}{3}$$
$$A = \frac{14}{3} - \ln 4 = \frac{14}{3} - 2 \ln 2$$

Question 22

Find the area of the region bounded by $y = \cos x$ and $y = \sec^2 x$, for $-\pi/4 \leq x \leq \pi/4$.

Solution On the interval $[-\pi/4, \pi/4]$, $\cos x \geq \sec^2 x$. (Correction: $\sec^2 x \geq 1$ and $\cos x \leq 1$, so $\cos x$ is the *lower* function on this central interval where $\sec x \geq 1$. The curves intersect when $\cos x = 1$ or $\cos x = 0$, so they actually intersect at $x = 0$. The proper upper/lower bounds depend on geometry. Assuming the region enclosed by the intersection points nearest the y-axis, the curves intersect when $\cos x = \sec^2 x$, i.e., $\cos^3 x = 1$, or $x = 0$. Given the endpoints, we must determine the upper/lower relationship on $[-\pi/4, \pi/4]$. $\sec^2 x$ is generally above $\cos x$ except possibly at points of intersection. Since $0 \leq x \leq \pi/4$, $\cos x \geq 0$ and $1 \leq \sec^2 x \leq 2$. At $x = 0$, $\cos 0 = 1$ and $\sec^2 0 = 1$. Near $x = 0$, $\sec^2 x > \cos x$. Thus, $y = \sec^2 x$ is the upper curve.)

$$A = \int_{-\pi/4}^{\pi/4} (\sec^2 x - \cos x) dx = [\tan x - \sin x]_{-\pi/4}^{\pi/4}$$
$$A = \left(1 - \frac{\sqrt{2}}{2}\right) - \left(-1 - \left(-\frac{\sqrt{2}}{2}\right)\right) = \left(1 - \frac{\sqrt{2}}{2}\right) + \left(1 - \frac{\sqrt{2}}{2}\right)$$
$$A = 2 - \sqrt{2}$$

Question 23

Find the area of the region bounded by $x = y^2$ and $x = 2 - y^2$.

Solution We integrate with respect to y . The curves intersect when $y^2 = 2 - y^2$, so $2y^2 = 2$, or $y = \pm 1$. The right boundary is $x = 2 - y^2$ and the left boundary is $x = y^2$.

$$A = \int_{-1}^1 ((2 - y^2) - y^2) dy = \int_{-1}^1 (2 - 2y^2) dy$$
$$A = 2 \left[2y - \frac{2y^3}{3} \right]_0^1 = 2 \left(2 - \frac{2}{3} \right) = 2 \left(\frac{4}{3} \right) = \frac{8}{3}$$

Question 24

Find the area of the region bounded by $y^2 = x$ and $x - y = 2$.

Solution We integrate with respect to y . The functions are $x = y^2$ and $x = y + 2$. Intersection: $y^2 = y + 2$, so $y^2 - y - 2 = 0$, or $(y - 2)(y + 1) = 0$. Intersections are at $y = -1$ and $y = 2$. The right boundary is $x = y + 2$ and the left boundary is $x = y^2$.

$$A = \int_{-1}^2 ((y + 2) - y^2) dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2$$
$$A = \left(\frac{4}{2} + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 - \frac{(-1)}{3} \right)$$
$$A = \left(6 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} - \left(-\frac{7}{6} \right) = \frac{20 + 7}{6} = \frac{27}{6} = \frac{9}{2}$$

Question 25

Find the area of the region bounded by $y = x^2$ and $y = 2x$.

Solution (Note: This is identical to Question 5. Repeating the unique method for completeness.) The curves intersect at $x = 0$ and $x = 2$. Upper curve is $y = 2x$.

$$A = \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = \frac{4}{3}$$

Question 26

Find the area of the region bounded by $y = x^3$ and $y = x$.

Solution (Note: This is identical to Question 8. Repeating the unique method for completeness.) The curves intersect at $x = -1$, $x = 0$, and $x = 1$. Upper curve is $y = x$ on $[2]$.

$$A = 2 \int_0^1 (x - x^3) dx = \frac{1}{2}$$

Question 27

Find the area of the region bounded by $x = 1 - y^4$ and $x = y^2 - 1$.

Solution We integrate with respect to y . Intersection points are $y = \pm 1$. The right boundary is $x = 1 - y^4$.

$$A = \int_{-1}^1 ((1 - y^4) - (y^2 - 1)) dy = \int_{-1}^1 (2 - y^2 - y^4) dy$$

Since the integrand is even:

$$A = 2 \left[2y - \frac{y^3}{3} - \frac{y^5}{5} \right]_0^1 = 2 \left(2 - \frac{1}{3} - \frac{1}{5} \right) = 2 \left(\frac{22}{15} \right) = \frac{44}{15}$$

Question 28

Find the area of the region bounded by $y = x^3$ and $y = x$.

Solution (Note: This is identical to Question 8 and 16. Repeating the unique method for completeness.) The curves intersect at $x = -1$, $x = 0$, and $x = 1$. The area is $1/2$.

Question 29

Find the area of the region bounded by $y = x^2$ and $y = 4$.

Solution The curves intersect when $x^2 = 4$, so $x = \pm 2$. The upper boundary is $y = 4$.

$$A = \int_{-2}^2 (4 - x^2) dx = 2 \int_0^2 (4 - x^2) dx$$

$$A = 2 \left[4x - \frac{x^3}{3} \right]_0^2 = 2 \left(8 - \frac{8}{3} \right) = 2 \left(\frac{16}{3} \right) = \frac{32}{3}$$

Question 30

Find the area of the region bounded by $y = x$, $y = 1/x^2$, and $x = 2$.

Solution The intersection points are $y = x$ and $y = 1/x^2$. Setting them equal gives $x = 1/x^2$, or $x^3 = 1$, so $x = 1$. The limits are $x = 1$ to $x = 2$. On $[1, 2]$, the upper curve is $y = x$ and the lower curve is $y = 1/x^2$.

$$A = \int_1^2 \left(x - \frac{1}{x^2} \right) dx = \left[\frac{x^2}{2} + \frac{1}{x} \right]_1^2$$

$$A = \left(\frac{4}{2} + \frac{1}{2} \right) - \left(\frac{1}{2} + \frac{1}{1} \right) = 2.5 - 1.5 = 1$$

Trigonometric Substitution

Question 1

$$\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} dx$$

Solution Let

$$x = 4 \sin \theta, \quad dx = 4 \cos \theta d\theta.$$

Then

$$\sqrt{16 - x^2} = 4 \cos \theta.$$

Changing the limits,

$$x = 0 \Rightarrow \theta = 0, \quad x = 2\sqrt{3} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2}, \quad \theta = \frac{\pi}{3}.$$

Substituting,

$$\begin{aligned} \int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} dx &= \int_0^{\pi/3} \frac{64 \sin^3 \theta}{4 \cos \theta} (4 \cos \theta) d\theta \\ &= 64 \int_0^{\pi/3} \sin^3 \theta d\theta. \end{aligned}$$

Write $\sin^3 \theta = \sin \theta (1 - \cos^2 \theta)$ and let $u = \cos \theta$, $du = -\sin \theta d\theta$. Then

$$\begin{aligned} 64 \int (\cos^2 \theta - 1)(-\sin \theta) d\theta &= 64 \int (1 - u^2) du \\ &= 64 \left(u - \frac{u^3}{3} \right) \Big|_{1/2}^1 = 64 \left(\frac{2}{3} - \frac{11}{24} \right) = 14. \end{aligned}$$

Question 2

$$\int_{\sqrt{2}}^2 \frac{1}{t^3 \sqrt{t^2 - 1}} dt$$

Solution Let

$$t = \sec \theta, \quad dt = \sec \theta \tan \theta d\theta.$$

Then

$$\sqrt{t^2 - 1} = \tan \theta.$$

Changing the limits,

$$t = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}, \quad t = 2 \Rightarrow \theta = \frac{\pi}{3}.$$

Substituting,

$$\begin{aligned} \int_{\sqrt{2}}^2 \frac{1}{t^3 \sqrt{t^2 - 1}} dt &= \int_{\pi/4}^{\pi/3} \frac{1}{\sec^3 \theta \tan \theta} (\sec \theta \tan \theta) d\theta \\ &= \int_{\pi/4}^{\pi/3} \cos^2 \theta d\theta. \end{aligned}$$

Using $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$,

$$\int \cos^2 \theta d\theta = \frac{1}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right).$$

Thus,

$$\frac{1}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_{\pi/4}^{\pi/3} = \frac{\pi}{24}.$$

Question 3

$$\int_0^2 x^3 \sqrt{x^2 + 4} dx$$

Solution Let

$$x = 2 \tan \theta, \quad dx = 2 \sec^2 \theta d\theta.$$

Then

$$\sqrt{x^2 + 4} = 2 \sec \theta.$$

Changing the limits,

$$x = 0 \Rightarrow \theta = 0, \quad x = 2 \Rightarrow \theta = \frac{\pi}{4}.$$

Substituting,

$$\begin{aligned} \int_0^2 x^3 \sqrt{x^2 + 4} \, dx &= \int_0^{\pi/4} (8 \tan^3 \theta)(2 \sec \theta)(2 \sec^2 \theta) \, d\theta \\ &= 32 \int_0^{\pi/4} \tan^3 \theta \sec^3 \theta \, d\theta. \end{aligned}$$

Write $\tan^3 \theta = \tan \theta (\sec^2 \theta - 1)$ and let $u = \sec \theta$, $du = \sec \theta \tan \theta \, d\theta$. Then

$$\begin{aligned} 32 \int (u^2 - 1)u^2 \, du &= 32 \int (u^4 - u^2) \, du \\ &= 32 \left(\frac{u^5}{5} - \frac{u^3}{3} \right) \bigg|_1^{\sqrt{2}} = \frac{256}{15}. \end{aligned}$$

Question 4

$$\int \frac{1}{x^2 \sqrt{25 - x^2}} \, dx$$

Solution Let

$$x = 5 \sin \theta, \quad dx = 5 \cos \theta \, d\theta.$$

Then

$$\sqrt{25 - x^2} = 5 \cos \theta.$$

Substituting,

$$\begin{aligned} \int \frac{1}{25 \sin^2 \theta \cdot 5 \cos \theta} (5 \cos \theta) \, d\theta &= \frac{1}{25} \int \csc^2 \theta \, d\theta \\ &= -\frac{1}{25} \cot \theta + C. \end{aligned}$$

From the triangle,

$$\cot \theta = \frac{\sqrt{25 - x^2}}{x}.$$

Thus,

$$\int \frac{1}{x^2 \sqrt{25 - x^2}} \, dx = -\frac{1}{25} \frac{\sqrt{25 - x^2}}{x} + C.$$

Question 5

$$\int \frac{\sqrt{x^2 - a^2}}{x^4} \, dx$$

Solution Let

$$x = a \sec \theta, \quad dx = a \sec \theta \tan \theta \, d\theta.$$

Then

$$\sqrt{x^2 - a^2} = a \tan \theta.$$

Substituting,

$$\int \frac{a \tan \theta}{a^4 \sec^4 \theta} (a \sec \theta \tan \theta) \, d\theta = \frac{1}{a^2} \int \tan^2 \theta \cos^3 \theta \, d\theta.$$

Using $\tan^2 \theta = \sec^2 \theta - 1$,

$$\frac{1}{a^2} \int (\sec^2 \theta - 1) \cos^3 \theta \, d\theta = \frac{1}{a^2} \int (\cos \theta - \cos^3 \theta) \, d\theta.$$

Thus,

$$\frac{1}{a^2} \left(\sin \theta - \frac{1}{3} \sin^3 \theta \right) + C.$$

From the triangle,

$$\sin \theta = \frac{\sqrt{x^2 - a^2}}{x}.$$

Hence,

$$\int \frac{\sqrt{x^2 - a^2}}{x^4} dx = \frac{1}{a^2} \left(\frac{\sqrt{x^2 - a^2}}{x} - \frac{1}{3} \left(\frac{\sqrt{x^2 - a^2}}{x} \right)^3 \right) + C.$$

Question 6

$$\int \frac{dx}{\sqrt{x^2 + 16}}$$

Solution Let

$$x = 4 \tan \theta, \quad dx = 4 \sec^2 \theta d\theta.$$

Then

$$\sqrt{x^2 + 16} = 4 \sec \theta.$$

Substituting,

$$\begin{aligned} \int \frac{4 \sec^2 \theta}{4 \sec \theta} d\theta &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C. \end{aligned}$$

Returning to x ,

$$\sec \theta = \frac{\sqrt{x^2 + 16}}{4}, \quad \tan \theta = \frac{x}{4}.$$

Thus,

$$\int \frac{dx}{\sqrt{x^2 + 16}} = \ln \left| x + \sqrt{x^2 + 16} \right| + C.$$

Question 7

$$\int \frac{t^5}{\sqrt{t^2 + 2}} dt$$

Solution Let

$$t = \sqrt{2} \tan \theta, \quad dt = \sqrt{2} \sec^2 \theta d\theta.$$

Then

$$\sqrt{t^2 + 2} = \sqrt{2} \sec \theta.$$

Substituting,

$$\int \frac{(\sqrt{2} \tan \theta)^5}{\sqrt{2} \sec \theta} (\sqrt{2} \sec^2 \theta) d\theta = 8 \int \tan^5 \theta \sec \theta d\theta.$$

Write $\tan^5 \theta = \tan^4 \theta \tan \theta = (\sec^2 \theta - 1)^2 \tan \theta$ and let $u = \sec \theta$, $du = \sec \theta \tan \theta d\theta$. Then

$$\begin{aligned} 8 \int (u^2 - 1)^2 du &= 8 \int (u^4 - 2u^2 + 1) du \\ &= 8 \left(\frac{u^5}{5} - \frac{2u^3}{3} + u \right) + C. \end{aligned}$$

Returning to t ,

$$u = \sec \theta = \frac{\sqrt{t^2 + 2}}{\sqrt{2}}.$$

Thus,

$$\int \frac{t^5}{\sqrt{t^2 + 2}} dt = 8 \left(\frac{1}{5} \left(\frac{\sqrt{t^2 + 2}}{\sqrt{2}} \right)^5 - \frac{2}{3} \left(\frac{\sqrt{t^2 + 2}}{\sqrt{2}} \right)^3 + \frac{\sqrt{t^2 + 2}}{\sqrt{2}} \right) + C.$$

Question 8

$$\int \sqrt{1-4x^2} \, dx$$

Solution Let $2x = \sin \theta$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then

$$x = \frac{1}{2} \sin \theta, \quad dx = \frac{1}{2} \cos \theta \, d\theta,$$

and

$$\sqrt{1-4x^2} = \sqrt{1-(2x)^2} = \cos \theta.$$

Thus

$$\begin{aligned} \int \sqrt{1-4x^2} \, dx &= \int \cos \theta \left(\frac{1}{2} \cos \theta \right) d\theta = \frac{1}{4} \int (1 + \cos 2\theta) \, d\theta \\ &= \frac{1}{4} \left(\theta + \frac{1}{2} \sin 2\theta \right) + C = \frac{1}{4} (\theta + \sin \theta \cos \theta) + C \\ &= \frac{1}{4} \left[\sin^{-1}(2x) + 2x \sqrt{1-4x^2} \right] + C. \end{aligned}$$

Question 9

$$\int_0^1 x \sqrt{x^2+4} \, dx$$

Solution

$$\begin{aligned} &= \frac{1}{4} \int_4^5 u^{1/2} \left(\frac{1}{2} du \right) \quad [u = x^2 + 4, \, du = 2x \, dx] \\ &= \frac{1}{2} \cdot \frac{2}{3} \left[u^{3/2} \right]_4^5 = \frac{1}{3} (5\sqrt{5} - 8). \end{aligned}$$

Question 10

$$\int \frac{\sqrt{x^2-9}}{x^3} \, dx$$

Solution Let $x = 3 \sec \theta$, where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$. Then

$$dx = 3 \sec \theta \tan \theta \, d\theta \quad \text{and} \quad \sqrt{x^2-9} = 3 \tan \theta,$$

so

$$\begin{aligned} \int \frac{\sqrt{x^2-9}}{x^3} \, dx &= \int \frac{3 \tan \theta}{27 \sec^3 \theta} (3 \sec \theta \tan \theta) \, d\theta = \frac{1}{3} \int \frac{\tan^2 \theta}{\sec^2 \theta} \, d\theta \\ &= \frac{1}{3} \int \sin^2 \theta \, d\theta = \frac{1}{3} \int \frac{1}{2} (1 - \cos 2\theta) \, d\theta \\ &= \frac{1}{6} \theta - \frac{1}{12} \sin 2\theta + C = \frac{1}{6} \theta - \frac{1}{6} \sin \theta \cos \theta + C \\ &= \frac{1}{6} \sec^{-1} \left(\frac{x}{3} \right) - \frac{1}{6} \frac{\sqrt{x^2-9}}{x} \frac{3}{x} + C = \frac{1}{6} \sec^{-1} \left(\frac{x}{3} \right) - \frac{\sqrt{x^2-9}}{2x^2} + C. \end{aligned}$$

Question 11

$$\int \frac{du}{u\sqrt{5-u^2}}$$

Solution Let $u = \sqrt{5} \sin \theta$, so $du = \sqrt{5} \cos \theta \, d\theta$. Then

$$\begin{aligned} \int \frac{du}{u\sqrt{5-u^2}} &= \int \frac{\sqrt{5} \cos \theta}{\sqrt{5} \sin \theta \cdot \sqrt{5} \cos \theta} \, d\theta = \frac{1}{\sqrt{5}} \int \csc \theta \, d\theta \\ &= \frac{1}{\sqrt{5}} \ln |\sec \theta - \cot \theta| + C \quad [\text{By Exercise 2.39}] \\ &= \frac{1}{\sqrt{5}} \ln \left| \frac{\sqrt{5} - \sqrt{5-u^2}}{u} \right| + C = \frac{1}{\sqrt{5}} \ln \left| \frac{\sqrt{5-u^2}}{u} \right| + C. \end{aligned}$$

Question 12

$$\int \frac{x^2}{(a^2 - x^2)^{3/2}} dx$$

Solution Let $x = a \sin \theta$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then $dx = a \cos \theta d\theta$ and

$$\begin{aligned} \int \frac{x^2}{(a^2 - x^2)^{3/2}} dx &= \int \frac{a^2 \sin^2 \theta \cdot a \cos \theta d\theta}{a^3 \cos^3 \theta} = \int \tan^2 \theta d\theta \\ &= \int (\sec^2 \theta - 1) d\theta = \tan \theta - \theta + C \\ &= \frac{x}{\sqrt{a^2 - x^2}} - \sin^{-1}\left(\frac{x}{a}\right) + C. \end{aligned}$$

Question 13

$$\int \frac{dx}{x^2 \sqrt{16x^2 - 9}}$$

Solution Let $4x = 3 \sec \theta$, where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$. Then

$$dx = \frac{3}{4} \sec \theta \tan \theta d\theta \quad \text{and} \quad \sqrt{16x^2 - 9} = 3 \tan \theta,$$

so

$$\begin{aligned} \int \frac{dx}{x^2 \sqrt{16x^2 - 9}} &= \int \frac{\frac{3}{4} \sec \theta \tan \theta}{\left(\frac{3}{4} \sec \theta\right)^2 \cdot 3 \tan \theta} d\theta = \frac{4}{9} \int \cos \theta d\theta \\ &= \frac{4}{9} \sin \theta + C = \frac{4}{9} \cdot \frac{\sqrt{16x^2 - 9}}{4x} + C = \frac{\sqrt{16x^2 - 9}}{9x} + C. \end{aligned}$$

Question 14

$$\int \frac{x}{\sqrt{x^2 - 7}} dx$$

Solution Let $u = x^2 - 7$, so $du = 2x dx$. Then

$$\int \frac{x}{\sqrt{x^2 - 7}} dx = \frac{1}{2} \int \frac{du}{\sqrt{u}} = \frac{1}{2} \cdot 2\sqrt{u} + C = \sqrt{x^2 - 7} + C.$$

Question 15

$$\int \frac{dx}{[(ax)^2 - b^2]^{3/2}}$$

Solution Let $ax = b \sec \theta$, so $(ax)^2 = b^2 \sec^2 \theta = (ax)^2 - b^2 = b^2(\sec^2 \theta - 1) = b^2 \tan^2 \theta$. So

$$\sqrt{(ax)^2 - b^2} = b \tan \theta, \quad dx = \frac{b}{a} \sec \theta \tan \theta d\theta,$$

and

$$\begin{aligned} \int \frac{dx}{[(ax)^2 - b^2]^{3/2}} &= \int \frac{\frac{b}{a} \sec \theta \tan \theta}{b^3 \tan^3 \theta} d\theta = \frac{1}{ab^2} \int \frac{\sec \theta}{\tan^2 \theta} d\theta \\ &= \frac{1}{ab^2} \int \csc \theta \cot \theta d\theta = -\frac{1}{ab^2} \csc \theta + C \\ &= -\frac{x}{b^2 \sqrt{(ax)^2 - b^2}} + C. \end{aligned}$$

Question 16

$$\int \frac{\sqrt{1+x^2}}{x} dx$$

Solution Let $x = \tan \theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = \sec^2 \theta d\theta$ and $\sqrt{1+x^2} = \sec \theta$, so

$$\begin{aligned} \int \frac{\sqrt{1+x^2}}{x} dx &= \int \frac{\sec \theta}{\tan \theta} \sec^2 \theta d\theta = \int (\csc \theta + \sec \theta \tan \theta) d\theta \\ &= \ln |\csc \theta - \cot \theta| + \sec \theta + C \quad [\text{by Exercise 8.39}] \\ &= \ln \left| \frac{\sqrt{1+x^2}-1}{x} \right| + \sqrt{1+x^2} + C. \end{aligned}$$

Question 17

$$\int \frac{t}{\sqrt{25-t^2}} dt$$

Solution Let $t = 5 \sin \theta$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then $dt = 5 \cos \theta d\theta$ and $\sqrt{25-t^2} = 5 \cos \theta$, so

$$\begin{aligned} \int \frac{t}{\sqrt{25-t^2}} dt &= \int \frac{5 \sin \theta}{5 \cos \theta} 5 \cos \theta d\theta = 5 \int \sin \theta d\theta \\ &= -5 \cos \theta + C = -\sqrt{25-t^2} + C. \end{aligned}$$

Or: Let $u = 25 - t^2$, so $du = -2t dt$.

Question 18

$$\int_0^{2/3} x^3 \sqrt{4-9x^2} dx$$

Solution Let $u = 4 - 9x^2$, so $du = -18x dx$ and $x^2 = \frac{1}{9}(4-u)$. Then

$$\begin{aligned} \int_0^{2/3} x^3 \sqrt{4-9x^2} dx &= \int_4^0 \frac{1}{9}(4-u)x \left(-\frac{1}{18}\right) \sqrt{u} du \\ &= \frac{1}{162} \int_0^4 (4u^{1/2} - u^{3/2}) du = \frac{1}{162} \left[\frac{8}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^4 \\ &= \frac{1}{162} \left(\frac{64}{3} - \frac{64}{5} \right) = \frac{64}{1215}. \end{aligned}$$

Or: Let $3x = 2 \sin \theta$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

Question 19

$$\int_0^1 \sqrt{x^2+1} dx$$

Solution Let $x = \tan \theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = \sec^2 \theta d\theta$, $\sqrt{x^2+1} = \sec \theta$, and $x = 0 \Rightarrow \theta = 0$, $x = 1 \Rightarrow \theta = \frac{\pi}{4}$, so

$$\begin{aligned} \int_0^1 \sqrt{x^2+1} dx &= \int_0^{\pi/4} \sec^3 \theta d\theta \\ &= \frac{1}{2} [\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|]_0^{\pi/4} \quad [\text{by Example 8.28}] \\ &= \frac{1}{2} [\sqrt{2} + \ln(1 + \sqrt{2})]. \end{aligned}$$

Integration of Rational Functions by Partial Fractions

Question 1

Evaluate $\int \frac{1}{(t+4)(t-1)} dt$.

Solution We use partial fractions on the integrand $\frac{1}{(t+4)(t-1)} = \frac{A}{t+4} + \frac{B}{t-1}$. Multiplying by $(t+4)(t-1)$ gives $1 = A(t-1) + B(t+4)$. Setting $t = 1 \implies 1 = 5B \implies B = 1/5$. Setting $t = -4 \implies 1 = -5A \implies A = -1/5$.

$$\begin{aligned} \int \left(\frac{-1/5}{t+4} + \frac{1/5}{t-1} \right) dt &= \frac{1}{5} \int \left(\frac{1}{t-1} - \frac{1}{t+4} \right) dt \\ &= \frac{1}{5} (\ln |t-1| - \ln |t+4|) + C = \frac{1}{5} \ln \left| \frac{t-1}{t+4} \right| + C \quad [1] \end{aligned}$$

Question 2

Evaluate $\int_2^3 \frac{1}{x^2-1} dx$.

Solution We use partial fractions on the integrand $\frac{1}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$. Multiplying by $(x+1)(x-1)$ gives $1 = A(x-1) + B(x+1)$. Setting $x = 1 \implies 1 = 2B \implies B = 1/2$. Setting $x = -1 \implies 1 = -2A \implies A = -1/2$.

$$\begin{aligned} \int_2^3 \left(\frac{-1/2}{x+1} + \frac{1/2}{x-1} \right) dx &= \frac{1}{2} [\ln |x-1| - \ln |x+1|]_2^3 \\ &= \frac{1}{2} \left[\ln \left| \frac{x-1}{x+1} \right| \right]_2^3 = \frac{1}{2} \left(\ln \left(\frac{3-1}{3+1} \right) - \ln \left(\frac{2-1}{2+1} \right) \right) \\ &= \frac{1}{2} \left(\ln \left(\frac{2}{4} \right) - \ln \left(\frac{1}{3} \right) \right) = \frac{1}{2} \left(\ln \left(\frac{1}{2} \right) - \ln \left(\frac{1}{3} \right) \right) \\ &= \frac{1}{2} \ln \left(\frac{3}{2} \right) \quad [1] \end{aligned}$$

Question 3

Evaluate $\int_0^1 \frac{x-1}{x^2+3x+2} dx$.

Solution The denominator factors as $(x+1)(x+2)$. Partial fraction decomposition: $\frac{x-1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$. Multiplying by $(x+1)(x+2)$ gives $x-1 = A(x+2) + B(x+1)$. Setting $x = -1 \implies -2 = A(1) \implies A = -2$. Setting $x = -2 \implies -3 = B(-1) \implies B = 3$.

$$\begin{aligned} \int_0^1 \left(\frac{-2}{x+1} + \frac{3}{x+2} \right) dx &= [-2 \ln |x+1| + 3 \ln |x+2|]_0^1 \\ &= (-2 \ln(2) + 3 \ln(3)) - (-2 \ln(1) + 3 \ln(2)) \\ &= 3 \ln 3 - 5 \ln 2 = \ln \left(\frac{27}{32} \right) \quad [2] \end{aligned}$$

Question 4

Evaluate $\int \frac{x^3-4x-10}{x^2-x-6} dx$.

Solution Since the degree of the numerator (3) is greater than the degree of the denominator (2), we first perform long division:

$$\frac{x^3-4x-10}{x^2-x-6} = x+1 + \frac{3x-4}{x^2-x-6}$$

The rational part is $\frac{3x-4}{(x-3)(x+2)}$. Partial fraction decomposition: $\frac{3x-4}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. $3x-4 = A(x+2) + B(x-3)$. Setting $x=3 \Rightarrow 5=5A \Rightarrow A=1$. Setting $x=-2 \Rightarrow -10=-5B \Rightarrow B=2$.

$$\begin{aligned} & \int \left(x+1 + \frac{1}{x-3} + \frac{2}{x+2} \right) dx \\ &= \frac{1}{2}x^2 + x + \ln|x-3| + 2\ln|x+2| + C \quad [3] \end{aligned}$$

Question 5

Evaluate $\int \frac{x^2+2x-1}{x^3-x} dx$.

Solution The denominator factors as $x(x^2-1) = x(x-1)(x+1)$. Decomposition: $\frac{x^2+2x-1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$. Multiplying by $x(x-1)(x+1)$ gives $x^2+2x-1 = A(x^2-1) + Bx(x+1) + Cx(x-1)$. Setting $x=0 \Rightarrow -1=-A \Rightarrow A=1$. Setting $x=1 \Rightarrow 2=2B \Rightarrow B=1$. Setting $x=-1 \Rightarrow -2=2C \Rightarrow C=-1$.

$$\begin{aligned} & \int \left(\frac{1}{x} + \frac{1}{x-1} - \frac{1}{x+1} \right) dx \\ &= \ln|x| + \ln|x-1| - \ln|x+1| + C = \ln \left| \frac{x(x-1)}{x+1} \right| + C \quad [4] \end{aligned}$$

Question 6

Evaluate $\int \frac{1}{(x+5)^2(x-1)} dx$.

Solution The decomposition form is $\frac{A}{x+5} + \frac{B}{(x+5)^2} + \frac{C}{x-1}$. Multiplying by the denominator gives $1 = A(x+5)(x-1) + B(x-1) + C(x+5)^2$. Setting $x=1 \Rightarrow 1=C(36) \Rightarrow C=1/36$. Setting $x=-5 \Rightarrow 1=B(-6) \Rightarrow B=-1/6$. Equating coefficients of x^2 : $0=A+C \Rightarrow A=-1/36$.

$$\begin{aligned} & \int \left(\frac{-1/36}{x+5} + \frac{-1/6}{(x+5)^2} + \frac{1/36}{x-1} \right) dx \\ &= \frac{1}{36} \ln|x-1| - \frac{1}{36} \ln|x+5| + \frac{1}{6(x+5)} + C \quad [5] \end{aligned}$$

Question 7

Evaluate $\int \frac{x^2}{(x-3)(x+2)^2} dx$.

Solution The decomposition form is $\frac{A}{x-3} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$. Multiplying by the denominator gives $x^2 = A(x+2)^2 + B(x-3)(x+2) + C(x-3)$. Setting $x=3 \Rightarrow 9=A(25) \Rightarrow A=9/25$. Setting $x=-2 \Rightarrow 4=C(-5) \Rightarrow C=-4/5$. Equating coefficients of x^2 : $1=A+B \Rightarrow B=1-9/25=16/25$.

$$\begin{aligned} & \int \left(\frac{9/25}{x-3} + \frac{16/25}{x+2} - \frac{4/5}{(x+2)^2} \right) dx \\ &= \frac{9}{25} \ln|x-3| + \frac{16}{25} \ln|x+2| + \frac{4}{5(x+2)} + C \quad [6] \end{aligned}$$

Question 8

Evaluate $\int \frac{10}{(x-1)(x^2+9)} dx$.

Solution The quadratic factor x^2+9 is irreducible. Decomposition: $\frac{10}{(x-1)(x^2+9)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+9}$. Multiplying by $(x-1)(x^2+9)$ gives $10 = A(x^2+9) + (Bx+C)(x-1)$. Setting $x=1 \Rightarrow 10=10A \Rightarrow A=1$. Equating coefficients of x^2 : $0=A+B \Rightarrow B=-1$. Equating constant terms: $10=9A-C \Rightarrow 10=9(1)-C \Rightarrow C=-1$.

$$\int \left(\frac{1}{x-1} + \frac{-x-1}{x^2+9} \right) dx = \int \frac{1}{x-1} dx - \int \frac{x}{x^2+9} dx - \int \frac{1}{x^2+9} dx$$

$$\begin{aligned}
&= \ln|x-1| - \frac{1}{2} \ln(x^2+9) - \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C \\
&= \ln|\mathbf{x}-1| - \frac{1}{2} \ln(\mathbf{x}^2+9) - \frac{1}{3} \arctan(\mathbf{x}/3) + \mathbf{C} \quad [7]
\end{aligned}$$

Question 9

Evaluate $\int \frac{x^2-2x-1}{(x-1)^2(x^2+1)} dx$.

Solution The decomposition requires terms for the repeated linear factor and the irreducible quadratic factor: $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1}$. Solving for the constants gives $A=1, B=-1, C=-1, D=1$.

$$\begin{aligned}
&\int \left(\frac{1}{x-1} - \frac{1}{(x-1)^2} + \frac{-x+1}{x^2+1} \right) dx \\
&= \int \frac{dx}{x-1} - \int \frac{dx}{(x-1)^2} - \frac{1}{2} \int \frac{2x}{x^2+1} dx + \int \frac{1}{x^2+1} dx \\
&= \ln|\mathbf{x}-1| + \frac{1}{\mathbf{x}-1} - \frac{1}{2} \ln(\mathbf{x}^2+1) + \arctan \mathbf{x} + \mathbf{C} \quad [7], [8]
\end{aligned}$$

Question 10

Evaluate $\int \frac{dx}{x\sqrt{x+1}}$.

Solution Let $u = \sqrt{x+1}$. Then $u^2 = x+1$, so $x = u^2-1$, and $dx = 2udu$.

$$\int \frac{2udu}{(u^2-1)u} = 2 \int \frac{du}{u^2-1}$$

Using partial fractions: $\frac{1}{u^2-1} = \frac{1/2}{u-1} - \frac{1/2}{u+1}$.

$$\begin{aligned}
&= 2 \cdot \frac{1}{2} \int \left(\frac{1}{u-1} - \frac{1}{u+1} \right) du = \ln|u-1| - \ln|u+1| + C \\
&= \ln \left| \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1} \right| + \mathbf{C} \quad [9]
\end{aligned}$$

Question 11

Evaluate $\int \frac{x}{x^4+1} dx$.

Solution This does not require partial fractions, but a simple substitution to make the denominator suitable for arctan integration. Let $u = x^2$. Then $du = 2xdx$, so $xdx = \frac{1}{2}du$.

$$\begin{aligned}
&\int \frac{1}{u^2+1} \frac{1}{2} du = \frac{1}{2} \arctan u + C \\
&= \frac{1}{2} \arctan(\mathbf{x}^2) + \mathbf{C} \quad [10]
\end{aligned}$$

Question 12

Evaluate $\int \frac{x^3}{x^3+1} dx$.

Solution Long division first: $\frac{x^3}{x^3+1} = 1 - \frac{1}{x^3+1}$. The remaining term has the factorable denominator $x^3+1 = (x+1)(x^2-x+1)$.

$$\int 1 dx - \int \frac{1}{(x+1)(x^2-x+1)} dx$$

The decomposition of the rational part leads to terms involving $\ln|x+1|$, $\ln(x^2-x+1)$, and $\arctan$ (after completing the square in the denominator).

$$= \mathbf{x} - \frac{1}{3} \ln|\mathbf{x}+1| + \frac{1}{6} \ln(\mathbf{x}^2-\mathbf{x}+1) + \frac{1}{\sqrt{3}} \arctan\left(\frac{2\mathbf{x}-1}{\sqrt{3}}\right) + \mathbf{C} \quad [10]$$

Question 13

Evaluate $\int \frac{dx}{x\sqrt{x+2}}$.

Solution Let $u = \sqrt{x+2}$. Then $x = u^2 - 2$, and $dx = 2udu$.

$$\int \frac{2udu}{(u^2-2)u} = 2 \int \frac{du}{u^2-2} = 2 \int \frac{du}{u^2 - (\sqrt{2})^2}$$

Using partial fractions or Formula 6 ($\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$):

$$\begin{aligned} &= 2 \left[\frac{1}{2\sqrt{2}} \ln \left| \frac{u - \sqrt{2}}{u + \sqrt{2}} \right| \right] + C \\ &= \frac{1}{\sqrt{2}} \ln \left| \frac{\sqrt{x+2} - \sqrt{2}}{\sqrt{x+2} + \sqrt{2}} \right| + C \quad [11] \end{aligned}$$

Question 14

Evaluate $\int \frac{dx}{\sqrt{x}(x-1)}$.

Solution Let $u = \sqrt{x}$. Then $u^2 = x$, and $dx = 2udu$.

$$\int \frac{2udu}{u(u^2-1)} = 2 \int \frac{du}{u^2-1}$$

Using partial fractions (as in Q39): $\frac{1}{u^2-1} = \frac{1/2}{u-1} - \frac{1/2}{u+1}$.

$$\begin{aligned} &= 2 \cdot \frac{1}{2} \int \left(\frac{1}{u-1} - \frac{1}{u+1} \right) du = \ln|u-1| - \ln|u+1| + C \\ &= \ln \left| \frac{\sqrt{x}-1}{\sqrt{x}+1} \right| + C \quad [12] \end{aligned}$$

Question 15

Evaluate $\int \frac{dx}{\sqrt{x^3+\sqrt{x}}}$.

Solution Rewrite the denominator: $\sqrt{x^3+\sqrt{x}} = x\sqrt{x}+\sqrt{x} = \sqrt{x}(x+1)$. Let $u = \sqrt{x}$. Then $u^2 = x$, and $dx = 2udu$.

$$\begin{aligned} \int \frac{2udu}{\sqrt{x}(x+1)} &= \int \frac{2udu}{u(u^2+1)} = 2 \int \frac{du}{u^2+1} \\ &= 2 \arctan u + C = 2 \arctan(\sqrt{x}) + C \quad [13] \end{aligned}$$

Areas

Question 1

Find the area of the region bounded by the given curves.

$$y = \frac{1}{x}, \quad y = \frac{1}{x^2}, \quad x = 2$$

Solution

$$\begin{aligned} A &= \int_1^2 \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \left[\ln x + \frac{1}{x} \right]_1^2 \\ &= (\ln 2 + \frac{1}{2}) - (\ln 1 + 1) = \ln 2 - \frac{1}{2} \approx 0.19. \end{aligned}$$

Question 2

Find the area of the region bounded by the given curves.

$$y = 1 + \sqrt{x}, \quad y = \frac{3+x}{3}$$

Solution

$$1 + \sqrt{x} = \frac{3+x}{3} \Rightarrow 1 + \frac{x}{3} = 1 + \sqrt{x} \Rightarrow \sqrt{x} = \frac{x}{3} \Rightarrow x = \frac{x^2}{9} \Rightarrow 9x - x^2 = 0 \Rightarrow x(9-x) = 0 \Rightarrow x = 0 \text{ or } 9, \text{ so}$$

$$\begin{aligned} A &= \int_0^9 \left[(1 + \sqrt{x}) - \frac{3+x}{3} \right] dx = \int_0^9 \left[(1 + \sqrt{x}) - \left(1 + \frac{x}{3} \right) \right] dx \\ &= \int_0^9 \left(\sqrt{x} - \frac{1}{3}x \right) dx = \left[\frac{2}{3}x^{3/2} - \frac{1}{6}x^2 \right]_0^9 = 18 - \frac{27}{2} = \frac{9}{2}. \end{aligned}$$

Question 3

Find the area of the region bounded by the given curves.

$$y = x^2, \quad y^2 = x$$

Solution

$$\begin{aligned} A &= \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3}x^{3/2} - \frac{1}{3}x^3 \right]_0^1 \\ &= \frac{2}{3} - \frac{1}{3} = \frac{1}{3}. \end{aligned}$$

Question 4

Find the area of the region bounded by the given curves.

$$y = x, \quad y = \sqrt[3]{x}$$

Solution

$$\begin{aligned} A &= \int_{-1}^1 (\sqrt[3]{x} - x) dx = \int_{-1}^0 (\sqrt[3]{x} - x) dx + \int_0^1 (\sqrt[3]{x} - x) dx \\ &= 2 \int_0^1 (x^{1/3} - x) dx \quad [\text{by symmetry}] \\ &= 2 \left[\frac{3}{4}x^{4/3} - \frac{1}{2}x^2 \right]_0^1 = 2 \left(\frac{3}{4} - \frac{1}{2} \right) = \frac{1}{2}. \end{aligned}$$

Question 5

Find the area of the region bounded by the given curves.

$$y = \frac{1}{x}, \quad y = \frac{1}{x^2}, \quad x = 2$$

Solution

$$\begin{aligned} A &= \int_1^2 \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \left[\ln x + \frac{1}{x} \right]_1^2 \\ &= (\ln 2 + \frac{1}{2}) - (\ln 1 + 1) = \ln 2 - \frac{1}{2} \approx 0.19. \end{aligned}$$

Question 6

Find the area of the region bounded by the given curves.

$$y = 1 + \sqrt{x}, \quad y = \frac{3+x}{3}$$

Solution

$$1 + \sqrt{x} = \frac{3+x}{3} \Rightarrow 1 + \frac{x}{3} = 1 + \sqrt{x} \Rightarrow \sqrt{x} = \frac{x}{3} \Rightarrow x = \frac{x^2}{9} \Rightarrow 9x - x^2 = 0 \Rightarrow x(9-x) = 0 \Rightarrow x = 0 \text{ or } 9, \text{ so}$$

$$\begin{aligned} A &= \int_0^9 \left[(1 + \sqrt{x}) - \frac{3+x}{3} \right] dx = \int_0^9 \left[(1 + \sqrt{x}) - \left(1 + \frac{x}{3} \right) \right] dx \\ &= \int_0^9 \left(\sqrt{x} - \frac{1}{3}x \right) dx = \left[\frac{2}{3}x^{3/2} - \frac{1}{6}x^2 \right]_0^9 = 18 - \frac{27}{2} = \frac{9}{2}. \end{aligned}$$

Question 7

Find the area of the region bounded by the given curves.

$$y = x^2, \quad y^2 = x$$

Solution

$$\begin{aligned} A &= \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3}x^{3/2} - \frac{1}{3}x^3 \right]_0^1 \\ &= \frac{2}{3} - \frac{1}{3} = \frac{1}{3}. \end{aligned}$$

Question 8

Find the area of the region bounded by the given curves.

$$y = x, \quad y = \sqrt[3]{x}$$

Solution

$$\begin{aligned} A &= \int_{-1}^1 (\sqrt[3]{x} - x) dx = \int_{-1}^0 (\sqrt[3]{x} - x) dx + \int_0^1 (\sqrt[3]{x} - x) dx \\ &= 2 \int_0^1 (x^{1/3} - x) dx \quad [\text{by symmetry}] \\ &= 2 \left[\frac{3}{4}x^{4/3} - \frac{1}{2}x^2 \right]_0^1 = 2 \left(\frac{3}{4} - \frac{1}{2} \right) = \frac{1}{2}. \end{aligned}$$

Question 9

Find the area of the region bounded by the given curves.

$$y = 12 - x^2, \quad y = x^2 - 6$$

Solution

$$\begin{aligned} 12 - x^2 &= x^2 - 6 \Rightarrow 2x^2 = 18 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3, \text{ so} \\ A &= \int_{-3}^3 [(12 - x^2) - (x^2 - 6)] dx = 2 \int_0^3 (18 - 2x^2) dx \quad [\text{by symmetry}] \\ &= 2 \left[18x - \frac{2}{3}x^3 \right]_0^3 = 2[(54 - 18) - 0] = 72. \end{aligned}$$

Question 10

Find the area of the region bounded by the given curves.

$$y = x^3 - x, \quad y = 3x$$

Solution

$$x^3 - x = 3x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = -2, 0, 2.$$

By symmetry,

$$\begin{aligned} A &= 2 \int_0^2 [3x - (x^3 - x)] dx = 2 \int_0^2 (4x - x^3) dx \\ &= 2 \left[2x^2 - \frac{1}{4}x^4 \right]_0^2 = 2(8 - 4) = 8. \end{aligned}$$

Question 11

Find the area of the region bounded by the given curves.

$$y = \sqrt{x}, \quad y = \frac{1}{2}x, \quad x = 9$$

Solution

$$\frac{1}{2}x = \sqrt{x} \Rightarrow \frac{1}{4}x^2 = x \Rightarrow x(x - 4) = 0 \Rightarrow x = 0 \text{ or } 4, \text{ so}$$

$$\begin{aligned} A &= \int_0^4 \left(\sqrt{x} - \frac{1}{2}x \right) dx + \int_4^9 \left(\frac{1}{2}x - \sqrt{x} \right) dx \\ &= \left[\frac{2}{3}x^{3/2} - \frac{1}{4}x^2 \right]_0^4 + \left[\frac{1}{4}x^2 - \frac{2}{3}x^{3/2} \right]_4^9 \\ &= \left(\frac{16}{3} - 4 \right) + \left(\frac{81}{4} - 18 \right) - \left(4 - \frac{16}{3} \right) = \frac{59}{12}. \end{aligned}$$

Question 12

Find the area of the region bounded by the given curves.

$$y = 8 - x^2, \quad y = x^2, \quad x = -3, \quad x = 3$$

Solution

$$\begin{aligned} A &= \int_{-3}^3 |(8 - x^2) - x^2| dx = 2 \int_0^3 |8 - 2x^2| dx \\ &= 2 \int_0^2 (8 - 2x^2) dx + 2 \int_2^3 (2x^2 - 8) dx \\ &= 2 \left[8x - \frac{2}{3}x^3 \right]_0^2 + 2 \left[\frac{2}{3}x^3 - 8x \right]_2^3 \\ &= 2 \left[\left(16 - \frac{16}{3} \right) - 0 \right] + 2 \left[(18 - 24) - \left(\frac{16}{3} - 16 \right) \right] \\ &= 32 - \frac{32}{3} + 20 - \frac{32}{3} = 52 - \frac{64}{3} = \frac{92}{3}. \end{aligned}$$

Question 13

Find the area of the region bounded by the curves

$$x = 2y^2, \quad x + y = 1.$$

Solution

$$\begin{aligned} 2y^2 = 1 - y &\Rightarrow 2y^2 + y - 1 = 0 \\ (2y - 1)(y + 1) = 0 &\Rightarrow y = \frac{1}{2} \text{ or } y = -1. \\ A = \int_{-1}^{1/2} [(1 - y) - 2y^2] dy &= \int_{-1}^{1/2} (1 - y - 2y^2) dy \\ &= \left[y - \frac{1}{2}y^2 - \frac{2}{3}y^3 \right]_{-1}^{1/2} \\ &= \left(\frac{1}{2} - \frac{1}{8} - \frac{1}{12} \right) - \left(-1 - \frac{1}{2} + \frac{2}{3} \right) = \frac{7}{24} - \left(-\frac{5}{6} \right) = \frac{27}{24} = \frac{9}{8}. \end{aligned}$$

Question 14

Find the area of the region bounded by the curves

$$4x + y^2 = 12, \quad x = y.$$

Solution

$$\begin{aligned} 4x + y^2 = 12 &\Rightarrow x = 3 - \frac{1}{4}y^2. \\ 3 - \frac{1}{4}y^2 = y &\Rightarrow y^2 + 4y - 12 = 0 \Rightarrow y = -6 \text{ or } y = 2. \\ A = \int_{-6}^2 \left[\left(3 - \frac{1}{4}y^2 \right) - y \right] dy &= \left[-\frac{1}{12}y^3 - \frac{1}{2}y^2 + 3y \right]_{-6}^2 \\ &= \left(-\frac{2}{3} - 2 + 6 \right) - (-18 - 18 - 18) = \frac{64}{3}. \end{aligned}$$

Question 15

Find the area of the region bounded by the curves

$$x = 1 - y^2, \quad x = y^2 - 1.$$

Solution

$$\begin{aligned} 1 - y^2 = y^2 - 1 &\Rightarrow y^2 = 1 \Rightarrow y = \pm 1. \\ A = \int_{-1}^1 [(1 - y^2) - (y^2 - 1)] dy &= \int_{-1}^1 2(1 - y^2) dy \\ &= 2 \int_{-1}^1 (1 - y^2) dy = 4 \int_0^1 (1 - y^2) dy \\ &= 4 \left[y - \frac{1}{3}y^3 \right]_0^1 = 4 \left(1 - \frac{1}{3} \right) = \frac{8}{3}. \end{aligned}$$

Volumes: Disk, Shell Methods

Find volume of the solid obtained by rotating the region bounded by the given curves about the specified line. Sketch the region and obtain the volume.

Question 1

$y = x^2$, $x = 1$, $y = 0$; about the x -axis.

Solution A cross section is a disk with radius x^2 , so its area is

$$A(x) = \pi(x^2)^2.$$

Thus,

$$V = \pi \int_0^1 x^4 dx = \pi \left[\frac{1}{5} x^5 \right]_0^1 = \pi \frac{1}{5}.$$

Question 2

$y = e^x$, $y = 0$, $x = 0$, $x = 1$; about the x -axis.

Solution A cross section is a disk with radius e^x , so its area is

$$A(x) = \pi(e^x)^2.$$

Thus,

$$V = \pi \int_0^1 e^{2x} dx = \pi \left[\frac{1}{2} e^{2x} \right]_0^1 = \frac{\pi}{2}(e^2 - 1).$$

Question 3

$y = \frac{1}{x}$, $x = 1$, $x = 2$, $y = 0$; about the x -axis.

Solution A cross section is a disk with radius $\frac{1}{x}$, so its area is

$$A(x) = \pi \left(\frac{1}{x} \right)^2.$$

Thus,

$$V = \pi \int_1^2 \frac{1}{x^2} dx = \pi \left[-\frac{1}{x} \right]_1^2 = \frac{\pi}{2}.$$

Question 4

$y = \sqrt{x-1}$, $x = 2$, $x = 5$, $y = 0$; about the x -axis.

Solution A cross section is a disk with radius $\sqrt{x-1}$, so its area is

$$A(x) = \pi(\sqrt{x-1})^2 = x - 1.$$

Thus,

$$V = \pi \int_2^5 (x-1) dx = \pi \left[\frac{1}{2} x^2 - x \right]_2^5 = \frac{15\pi}{2}.$$

Question 5

$y = x^2$, $0 \leq x \leq 2$, $y = 4$, $x = 0$; about the y -axis.

Solution A cross section is a disk with radius $\sqrt{y}$, so its area is

$$A(y) = y.$$

Thus,

$$V = \pi \int_0^4 y dy = \pi \left[\frac{1}{2} y^2 \right]_0^4 = 8\pi.$$

Question 6

$x = y - y^2$, $x = 0$; about the y -axis.

Solution A cross section is a disk with radius $y - y^2$, so its area is

$$A(y) = \pi(y - y^2)^2.$$

Thus,

$$\begin{aligned} V &= \pi \int_0^1 (y - y^2)^2 dy = \pi \int_0^1 (y^2 - 2y^3 + y^4) dy \\ &= \pi \left[\frac{1}{3}y^3 - \frac{1}{2}y^4 + \frac{1}{5}y^5 \right]_0^1 = \frac{\pi}{30}. \end{aligned}$$

Question 7

$y = x^2$, $y^2 = x$; about the x -axis.

Solution A cross section is a washer with outer radius $\sqrt{x}$ and inner radius x^2 , so its area is

$$A(x) = \pi(x - x^4).$$

Thus,

$$V = \int_0^1 \pi(x - x^4) dx = \pi \left[\frac{1}{2}x^2 - \frac{1}{5}x^5 \right]_0^1 = \frac{3\pi}{10}.$$

Question 8

$y = \sec x$, $y = 1$, $x = -1$, $x = 1$; about the x -axis.

Solution A cross section is a washer with outer radius $\sec x$ and inner radius 1, so its area is

$$A(x) = \pi(\sec^2 x - 1).$$

Thus,

$$V = \pi \int_{-1}^1 (\sec^2 x - 1) dx = \pi [\tan x - x]_{-1}^1 = 2\pi(\tan 1 - 1).$$

Question 9

$y^2 = x$, $x = 2y$; about the y -axis.

Solution A cross section is a washer with outer radius $2y$ and inner radius y^2 , so its area is

$$A(y) = \pi(4y^2 - y^4).$$

Thus,

$$V = \pi \int_0^2 (4y^2 - y^4) dy = \pi \left[\frac{4}{3}y^3 - \frac{1}{5}y^5 \right]_0^2 = \frac{64\pi}{15}.$$

Question 10

$y = x^{2/3}$, $x = 1$, $y = 0$; about the y -axis.

Solution A cross section is a washer with outer radius 1 and inner radius $y^{3/2}$, so its area is

$$A(y) = \pi((1)^2 - (y^{3/2})^2) = \pi(1 - y^3).$$

Thus,

$$V = \pi \int_0^1 (1 - y^3) dy = \pi \left[y - \frac{1}{4}y^4 \right]_0^1 = \frac{3\pi}{4}.$$

Question 11

$y = x$, $y = \sqrt{x}$; about $y = 1$.

Solution A cross section is a washer with outer radius $1 - x$ and inner radius $1 - \sqrt{x}$, so its area is

$$A(x) = \pi((1 - x)^2 - (1 - \sqrt{x})^2).$$

Thus,

$$V = \pi \int_0^1 [(1 - x)^2 - (1 - \sqrt{x})^2] dx = \pi/6.$$

Question 12

$y = x^2$, $y = 4$; about $y = 4$.

Solution A cross section is a disk with radius $4 - x^2$, so its area is

$$A(x) = \pi(4 - x^2)^2.$$

Thus,

$$V = \pi \int_0^2 (4 - x^2)^2 dx = \frac{512\pi}{15}.$$

Question 13

$y = x^4$, $y = 1$; about $y = 2$.

Solution A cross section is an annulus with inner radius 1 and outer radius $2 - x^4$, so its area is

$$A(x) = \pi((2 - x^4)^2 - (1)^2) = \pi(3 - 4x^4 + x^8).$$

Thus,

$$\begin{aligned} V &= \pi \int_{-1}^1 (3 - 4x^4 + x^8) dx = 2\pi \int_0^1 (3 - 4x^4 + x^8) dx \\ &= 2\pi \left[3x - \frac{4}{5}x^5 + \frac{1}{9}x^9 \right]_0^1 = \frac{208\pi}{45}. \end{aligned}$$

Question 14

$y = \frac{1}{x}$, $y = 0$, $x = 1$, $x = 3$; about $y = -1$.

Solution Thus,

$$\begin{aligned} V &= \pi \int_1^3 \left\{ \left(\frac{1}{x} + 1 \right)^2 - (1)^2 \right\} dx = \pi \int_1^3 \left(\frac{1}{x^2} + \frac{2}{x} \right) dx \\ &= \pi \left[-\frac{1}{x} + 2 \ln x \right]_1^3 = 2\pi \ln 3 + \frac{2\pi}{3}. \end{aligned}$$

Question 15

$x = y^2$, $x = 1$; about $x = 1$.

Solution

$$\begin{aligned} V &= \pi \int_{-1}^1 (1 - y^2)^2 dy = 2\pi \int_0^1 (1 - y^2)^2 dy = 2\pi \int_0^1 (1 - 2y^2 + y^4) dy \\ &= 2\pi \left[y - \frac{2}{3}y^3 + \frac{1}{5}y^5 \right]_0^1 = \frac{16\pi}{15}. \end{aligned}$$

Question 16

$y = x$, $y = \sqrt{x}$; about $x = 2$

Solution $y = \sqrt{x} \Rightarrow x = y^2$, so the outer radius is $2 - y^2$.

$$\begin{aligned} V &= \int_0^1 \pi [(2 - y^2)^2 - (2 - y)^2] dy = \pi \int_0^1 [(4 - 4y^2 + y^4) - (4 - 4y + y^2)] dy \\ &= \pi \int_0^1 (y^4 - 5y^2 + 4y) dy = \pi \left[\frac{1}{5}y^5 - \frac{5}{3}y^3 + 2y^2 \right]_0^1 = \pi \left(\frac{1}{5} - \frac{5}{3} + 2 \right) = \frac{8\pi}{15}. \end{aligned}$$

Question 17

$y = x^2$, $x = y^2$; about $x = -1$

Solution $y = x^2 \Rightarrow x = \sqrt{y}$ for $x \geq 0$. The outer radius is the distance from $x = -1$ to $x = \sqrt{y}$ and the inner radius is the distance from $x = -1$ to $x = y^2$.

$$\begin{aligned} V &= \int_0^1 \pi \{ [\sqrt{y} - (-1)]^2 - [y^2 - (-1)]^2 \} dy = \pi \int_0^1 [(\sqrt{y} + 1)^2 - (y^2 + 1)^2] dy \\ &= \pi \int_0^1 (y + 2\sqrt{y} + 1 - y^4 - 2y^2 - 1) dy = \pi \int_0^1 (y + 2\sqrt{y} - y^4 - 2y^2) dy \\ &= \pi \left[\frac{1}{2}y^2 + \frac{4}{3}y^{3/2} - \frac{1}{5}y^5 - \frac{2}{3}y^3 \right]_0^1 = \pi \left(\frac{1}{2} + \frac{4}{3} - \frac{1}{5} - \frac{2}{3} \right) = \frac{29\pi}{30}. \end{aligned}$$

Question 18

$y = x$, $y = 0$, $x = 2$, $x = 4$; about $x = 1$

Solution For $0 \leq y < 2$, a cross-section is an annulus with inner radius $2 - 1$ and outer radius $4 - 1$, the area of which is $A_1(y) = \pi(4 - 1)^2 - \pi(2 - 1)^2$. For $2 \leq y \leq 4$, a cross-section is an annulus with inner radius $y - 1$ and outer radius $4 - 1$, the area of which is $A_2(y) = \pi(4 - 1)^2 - \pi(y - 1)^2$.

$$\begin{aligned} V &= \int_0^2 A_1(y) dy + \int_2^4 A_2(y) dy = \int_0^2 \pi (3^2 - 1^2) dy + \int_2^4 \pi (3^2 - (y - 1)^2) dy \\ &= 16\pi + \pi \int_2^4 (9 - (y - 1)^2) dy = 16\pi + \pi \int_1^3 (9 - u^2) du \\ &= 16\pi + \pi \left[9u - \frac{1}{3}u^3 \right]_1^3 = 16\pi + \pi \left(18 - \frac{26}{3} \right) = 16\pi + \frac{28\pi}{3} = \frac{76\pi}{3}. \end{aligned}$$

(Equivalent simplified final value from the manual computation:

$$V = \pi \int_0^1 (2x^3 - x^6) dx = \pi \left[\frac{1}{2}x^4 - \frac{1}{7}x^7 \right]_0^1 = \frac{5\pi}{14}.$$

)

Set up, but do not evaluate, an integral for the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

Question 19

$y = \tan^3 x$, $y = 1$, $x = 0$; about $y = 1$

Solution

$$V = \pi \int_0^{\pi/4} (1 - \tan^3 x)^2 dx.$$

Question 20

$y = (x - 2)^4$, $8x - y = 16$; about $x = 10$

Solution $y = (x - 2)^4$ and $8x - y = 16$ intersect when $(x - 2)^4 = 8x - 16 = 8(x - 2)$, so $x - 2 = 0$ or $x - 2 = 2$, i.e. $x = 2$ or 4 .

$$y = (x - 2)^4 \Rightarrow x - 2 = \sqrt[4]{y} \Rightarrow x = 2 + \sqrt[4]{y} \quad (\text{since } x \geq 2), \quad 8x - y = 16 \Rightarrow x = \frac{1}{8}y + 2.$$

$$V = \pi \int_0^{16} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right) \right]^2 - [10 - (2 + \sqrt[4]{y})]^2 \right\} dy.$$

Question 21

$y = 0$, $y = \sin x$, $0 \leq x \leq \pi$; about $y = 1$

Solution

$$V = \pi \int_0^\pi [(1 - 0)^2 - (1 - \sin x)^2] dx = \pi \int_0^\pi [1^2 - (1 - \sin x)^2] dx.$$

Question 22

$y = 0$, $y = \sin x$, $0 \leq x \leq \pi$; about $y = -2$

Solution

$$V = \pi \int_0^\pi [(\sin x + 2)^2 - 2^2] dx.$$

Question 23

$x^2 - y^2 = 1$, $x = 3$; about $x = -2$

Solution

$$V = \pi \int_{-\sqrt{8}}^{\sqrt{8}} \left\{ [3 - (-2)]^2 - [\sqrt{y^2 + 1} - (-2)]^2 \right\} dy = \pi \int_{-2\sqrt{2}}^{2\sqrt{2}} \left[25 - (\sqrt{1 + y^2} + 2)^2 \right] dy.$$

Question 24

$2x + 3y = 6$, $(y - 1)^2 = 4 - x$; about $x = -5$

Solution Solve the equations for x :

$$(y - 1)^2 = 4 - x \iff x = 4 - (y - 1)^2, \quad 2x + 3y = 6 \iff x = 3 - \frac{3}{2}y.$$

The points of intersection of the two curves are $(3, 0)$ and $(-\frac{9}{4}, \frac{7}{2})$. Therefore,

$$\begin{aligned} V &= \pi \int_0^{7/2} \left\{ [4 - (y - 1)^2 - (-5)]^2 - \left[3 - \frac{3}{2}y - (-5) \right]^2 \right\} dy \\ &= \pi \int_0^{7/2} \left\{ [9 - (y - 1)^2]^2 - \left(8 - \frac{3}{2}y \right)^2 \right\} dy. \end{aligned}$$

Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the y -axis. Sketch the region and a typical shell.

Question 25

$y = \frac{1}{x}$, $y = 0$, $x = 1$, $x = 2$

Solution

$$V = 2\pi \int_1^2 x \left(\frac{1}{x} - 0 \right) dx = 2\pi \int_1^2 1 dx = 2\pi.$$

Question 26

$$y = x^2, y = 0, x = 1$$

Solution

$$V = 2\pi \int_0^1 x(x^2 - 0) dx = 2\pi \int_0^1 x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^1 = \frac{\pi}{2}.$$

Question 27

$$y = e^{-x^2}, y = 0, x = 0, x = 1$$

Solution

$$V = 2\pi \int_0^1 x(e^{-x^2} - 0) dx.$$

Let $u = x^2$, so $du = 2x dx$. Then

$$V = 2\pi \cdot \frac{1}{2} \int_0^1 e^{-u} du = \pi [-e^{-u}]_0^1 = \pi \left(1 - \frac{1}{e} \right).$$

Question 28

$$y = 3 + 2x - x^2, x + y = 3$$

Solution The curves intersect at $x = 0$ and $x = 3$. Using shells about the y -axis,

$$V = 2\pi \int_0^3 x[(3 + 2x - x^2) - (3 - x)] dx = 2\pi \int_0^3 x(3x - x^2) dx = 2\pi \int_0^3 (3x^2 - x^3) dx.$$

Thus,

$$V = 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^3 = 2\pi \left(27 - \frac{81}{4} \right) = \frac{27\pi}{2}.$$

Question 29

$$y = 4(x - 2)^2, y = x^2 - 4x + 7$$

Solution The curves intersect at $x = 1$ and $x = 3$. Using shells about the y -axis,

$$V = 2\pi \int_1^3 x[(x^2 - 4x + 7) - 4(x - 2)^2] dx = 2\pi \int_1^3 x(-3x^2 + 12x - 9) dx.$$

Hence,

$$V = 2\pi \int_1^3 (-3x^3 + 12x^2 - 9x) dx = 2\pi \left[-\frac{3}{4}x^4 + 4x^3 - \frac{9}{2}x^2 \right]_1^3 = 16\pi.$$

Use the method of cylindrical shells to find the volume of the solid obtained by rotating the region bounded by the given curves about the x -axis. Sketch the region and a typical shell.

Question 30

$$x = 1 + y^2, x = 0, y = 1, y = 2$$

Solution

$$V = 2\pi \int_1^2 y[(1 + y^2) - 0] dy = 2\pi \int_1^2 (y + y^3) dy = 2\pi \left[\frac{y^2}{2} + \frac{y^4}{4} \right]_1^2 = \frac{21\pi}{2}.$$

Question 31

$$x = \sqrt{y}, x = 0, y = 1$$

Solution

$$V = 2\pi \int_0^1 y(\sqrt{y} - 0) dy = 2\pi \int_0^1 y^{3/2} dy = 2\pi \left[\frac{2}{5} y^{5/2} \right]_0^1 = \frac{4\pi}{5}.$$

Question 32

$$y = x^3, \quad y = 8, \quad x = 0$$

Solution Write x as a function of y : $x = y^{1/3}$. Then

$$V = 2\pi \int_0^8 y(y^{1/3} - 0) dy = 2\pi \int_0^8 y^{4/3} dy = 2\pi \left[\frac{3}{7} y^{7/3} \right]_0^8 = 2\pi \cdot \frac{3}{7} \cdot 8^{7/3} = \frac{768\pi}{7}.$$

Question 33

$$x = 4y^2 - y^3, \quad x = 0$$

Solution The curves meet when $4y^2 - y^3 = 0$, i.e. $y = 0$ and $y = 4$. Thus,

$$V = 2\pi \int_0^4 y[(4y^2 - y^3) - 0] dy = 2\pi \int_0^4 (4y^3 - y^4) dy = 2\pi \left[y^4 - \frac{y^5}{5} \right]_0^4 = \frac{512\pi}{5}.$$

Question 34

$$y = 4x^2, \quad 2x + y = 6$$

Solution

$$\begin{aligned} 13. \text{ The curves intersect when } 4x^2 = 6 - 2x &\iff 2x^2 + x - 3 = 0 \iff (2x + 3)(x - 1) = 0 \\ &\iff x = -\frac{3}{2} \text{ or } 1. \end{aligned}$$

Solving the equations for x gives us $y = 4x^2 \Rightarrow x = \pm \frac{1}{2}\sqrt{y}$ and $2x + y = 6 \Rightarrow x = -\frac{1}{2}y + 3$.

$$\begin{aligned} V &= 2\pi \int_0^4 \left\{ y \left[\left(\frac{1}{2}\sqrt{y} \right) - \left(-\frac{1}{2}\sqrt{y} \right) \right] \right\} dy + 2\pi \int_4^9 \left\{ y \left[\left(-\frac{1}{2}y + 3 \right) - \left(-\frac{1}{2}\sqrt{y} \right) \right] \right\} dy \\ &= 2\pi \int_0^4 y\sqrt{y} dy + 2\pi \int_4^9 \left(-\frac{1}{2}y^2 + 3y + \frac{1}{2}y^{3/2} \right) dy \\ &= 2\pi \left[\frac{2}{5}y^{5/2} \right]_0^4 + 2\pi \left[-\frac{1}{6}y^3 + \frac{3}{2}y^2 + \frac{1}{5}y^{5/2} \right]_4^9 \\ &= 2\pi \left(\frac{2}{5} \cdot 32 \right) + 2\pi \left[\left(-\frac{243}{2} + \frac{243}{2} + \frac{243}{5} \right) - \left(-\frac{32}{3} + 24 + \frac{32}{5} \right) \right] \\ &= \frac{128}{5}\pi + 2\pi \left(\frac{433}{15} \right) = \frac{1250}{15}\pi = \frac{250}{3}\pi. \end{aligned}$$

Question 35

$$x + y = 3, \quad x = 4 - (y - 1)^2$$

Solution Here $x = 3 - y$ and $x = 4 - (y - 1)^2$. They intersect when $3 - y = 4 - (y - 1)^2$, which gives $y = 0$ and $y = 3$. Thus,

$$V = 2\pi \int_0^3 y[(4 - (y - 1)^2) - (3 - y)] dy = 2\pi \int_0^3 y(y^2 + 3y) dy = 2\pi \left[\frac{y^4}{4} + y^3 \right]_0^3 = \frac{27\pi}{2}.$$

Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the specified axis. Sketch the region and a typical shell.

Question 36

$$y = x^2, \quad y = 0, \quad x = 1, \quad x = 2; \text{ about } x = 1$$

Solution

$$V = \int_1^2 2\pi(x - 1)x^2 dx = 2\pi \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 \right]_1^2 = 2\pi \left[\left(4 - \frac{8}{3} \right) - \left(\frac{1}{4} - \frac{1}{3} \right) \right] = \frac{17\pi}{6}.$$

Question 37

$y = x^2$, $y = 0$, $x = -2$, $x = -1$; about the y -axis

Solution

$$V = \int_{-2}^{-1} 2\pi(-x)x^2 dx = 2\pi \left[-\frac{1}{4}x^4 \right]_{-2}^{-1} = 2\pi \left[\left(-\frac{1}{4} \right) - (-4) \right] = \frac{15\pi}{2}.$$

Question 38

$y = x^2$, $y = 0$, $x = 1$, $x = 2$; about $x = 4$

Solution

$$V = \int_1^2 2\pi(4-x)x^2 dx = 2\pi \left[\frac{4}{3}x^3 - \frac{1}{4}x^4 \right]_1^2 = 2\pi \left[\left(\frac{32}{3} - 4 \right) - \left(\frac{4}{3} - \frac{1}{4} \right) \right] = \frac{67\pi}{6}.$$

Question 39

$y = 8x - 2x^2$, $y = 4x - x^2$; about $x = -2$

Solution

$$\begin{aligned} V &= \int_0^4 2\pi[x - (-2)][(8x - 2x^2) - (4x - x^2)] dx = \int_0^4 2\pi(2+x)(4x - x^2) dx \\ &= 2\pi \int_0^4 (8x + 2x^2 - x^3) dx = 2\pi \left[4x^2 + \frac{2}{3}x^3 - \frac{1}{4}x^4 \right]_0^4 \\ &= 2\pi \left(64 + \frac{128}{3} - 64 \right) = \frac{256\pi}{3}. \end{aligned}$$

Question 40

$x = 1 + y^2$, $x = 5$, $y = 0$, $y = 2$; about $y = 3$

Solution

$$\begin{aligned} V &= \int_0^2 2\pi(3-y)(5-y^2-1) dy \\ &= \int_0^2 2\pi(12-4y-3y^2+y^3) dy = 2\pi \left[12y - 2y^2 - y^3 + \frac{1}{4}y^4 \right]_0^2 \\ &= 2\pi(24 - 8 - 8 + 4) = 24\pi. \end{aligned}$$

Question 41

$x = y^2$, $x = \sqrt{y}$; about $y = -1$

Solution

$$\begin{aligned} V &= \int_0^1 2\pi(y+1)(\sqrt{y}-y^2) dy = 2\pi \int_0^1 (y^{3/2} + y^{1/2} - y^3 - y^2) dy \\ &= 2\pi \left[\frac{2}{5}y^{5/2} + \frac{2}{3}y^{3/2} - \frac{1}{4}y^4 - \frac{1}{3}y^3 \right]_0^1 = 2\pi \left(\frac{2}{5} + \frac{2}{3} - \frac{1}{4} - \frac{1}{3} \right) = \frac{29\pi}{30}. \end{aligned}$$

Arc Length and Surface Area

Question 1

Find the length of the curve

$$y = \ln x, \quad 1 \leq x \leq \sqrt{3}.$$

Solution

$$\frac{dy}{dx} = \frac{1}{x}, \quad 1 + \left(\frac{dy}{dx} \right)^2 = 1 + \frac{1}{x^2} = \frac{1+x^2}{x^2}.$$

$$L = \int_1^{\sqrt{3}} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_1^{\sqrt{3}} \frac{\sqrt{1+x^2}}{x} dx.$$

Let $x = \sqrt{1+u^2}$, so that $dx = \frac{u}{\sqrt{1+u^2}} du$. Then

$$L = \int_{\sqrt{2}}^2 \left(1 + \frac{1}{u^2}\right)^{1/2} du = \left[u + \frac{1}{2} \ln \left|\frac{u-1}{u+1}\right|\right]_{\sqrt{2}}^2.$$

$$L = 2 + \ln(\sqrt{2}+1) - \frac{1}{2} \ln 3.$$

Question 2

Find the length of the curve

$$y = \cosh x, \quad 0 \leq x \leq 1.$$

Solution

$$\frac{dy}{dx} = \sinh x, \quad 1 + \sinh^2 x = \cosh^2 x.$$

$$L = \int_0^1 \cosh x dx = [\sinh x]_0^1 = \sinh 1 = \frac{1}{2}(e - e^{-1}).$$

Question 3

Find the length of the curve

$$y^2 = 4x, \quad 0 \leq y \leq 2.$$

Solution

$$x = \frac{1}{4}y^2, \quad \frac{dx}{dy} = \frac{1}{2}y.$$

$$L = \int_0^2 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_0^2 \sqrt{1 + \frac{1}{4}y^2} dy.$$

Let $u = \frac{1}{2}y$. Then

$$L = 2 \int_0^1 \sqrt{1+u^2} du = \left[u\sqrt{1+u^2} + \ln(u + \sqrt{1+u^2})\right]_0^1.$$

$$L = \sqrt{2} + \ln(1 + \sqrt{2}).$$

Question 4

Find the length of the curve

$$y = e^x, \quad 0 \leq x \leq 1.$$

Solution

$$\frac{dy}{dx} = e^x, \quad 1 + \left(\frac{dy}{dx}\right)^2 = 1 + e^{2x}.$$

$$L = \int_0^1 \sqrt{1 + e^{2x}} dx.$$

Let $u = e^x$. Then

$$L = \int_1^e \frac{\sqrt{1+u^2}}{u} du = \left[\sqrt{1+u^2} + \frac{1}{2} \ln \left|\frac{\sqrt{1+u^2}-1}{\sqrt{1+u^2}+1}\right|\right]_1^e.$$

$$L = \sqrt{1+e^2} - \sqrt{2} + \frac{1}{2} \ln \left(\frac{\sqrt{1+e^2}-1}{\sqrt{1+e^2}+1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}-1} \right).$$

Question 5

Find the length of the curve

$$y = \ln\left(\frac{e^x + 1}{e^x - 1}\right), \quad a \leq x \leq b, \quad a > 0.$$

Solution

$$\frac{dy}{dx} = \frac{e^x}{e^x + 1} - \frac{e^x}{e^x - 1} = -\frac{2e^x}{e^{2x} - 1}.$$

$$1 + \left(\frac{dy}{dx}\right)^2 = \frac{(e^{2x} + 1)^2}{(e^{2x} - 1)^2}.$$

$$L = \int_a^b \frac{e^{2x} + 1}{e^{2x} - 1} dx = [\ln(e^x - 1) - \ln(e^x + 1) + x]_a^b.$$

Question 6

Find the area of the surface obtained by rotating the curve about the x -axis.

$$y = x^3, \quad 0 \leq x \leq 2.$$

Solution

$$y' = 3x^2.$$

$$S = 2\pi \int_0^2 y \sqrt{1 + (y')^2} dx = 2\pi \int_0^2 x^3 \sqrt{1 + 9x^4} dx.$$

Let $u = 1 + 9x^4$, so $du = 36x^3 dx$. Then

$$S = \frac{2\pi}{36} \int_1^{145} \sqrt{u} du = \frac{\pi}{18} \left[\frac{2}{3} u^{3/2} \right]_1^{145} = \frac{\pi}{27} (145\sqrt{145} - 1).$$

Question 7

Find the area of the surface obtained by rotating the curve about the x -axis.

$$9x = y^2 + 18, \quad 2 \leq x \leq 6.$$

Solution The curve is symmetric about the x -axis, so we use only its top half,

$$y = 3\sqrt{x - 2}.$$

$$\frac{dy}{dx} = \frac{3}{2\sqrt{x - 2}}, \quad 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{9}{4(x - 2)}.$$

$$S = 2\pi \int_2^6 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 6\pi \int_2^6 \sqrt{x - 2} \sqrt{1 + \frac{9}{4(x - 2)}} dx.$$

$$= 6\pi \int_2^6 \sqrt{x + \frac{1}{4}} dx = 6\pi \left[\frac{2}{3} \left(x + \frac{1}{4}\right)^{3/2} \right]_2^6.$$

$$= 4\pi \left[\left(\frac{25}{4}\right)^{3/2} - \left(\frac{9}{4}\right)^{3/2} \right] = 4\pi \left(\frac{125}{8} - \frac{27}{8} \right) = 49\pi.$$

Question 8

Find the area of the surface obtained by rotating the curve about the x -axis.

$$y = \sqrt{x}, \quad 4 \leq x \leq 9.$$

Solution

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2\sqrt{x}}, & 1 + \left(\frac{dy}{dx}\right)^2 &= 1 + \frac{1}{4x}. \\ S &= 2\pi \int_4^9 \sqrt{x} \sqrt{1 + \frac{1}{4x}} dx = 2\pi \int_4^9 \sqrt{x + \frac{1}{4}} dx. \\ &= 2\pi \left[\frac{2}{3} \left(x + \frac{1}{4}\right)^{3/2} \right]_4^9 = \frac{4\pi}{3} \left[\left(\frac{37}{4}\right)^{3/2} - \left(\frac{17}{4}\right)^{3/2} \right]. \\ &= \frac{\pi}{6} (37\sqrt{37} - 17\sqrt{17}). \end{aligned}$$

Question 9

Find the area of the surface obtained by rotating the curve about the x -axis.

$$y = \cos 2x, \quad 0 \leq x \leq \frac{\pi}{6}.$$

Solution

$$\begin{aligned} \frac{dy}{dx} &= -2 \sin 2x, & 1 + \left(\frac{dy}{dx}\right)^2 &= 1 + 4 \sin^2 2x. \\ S &= 2\pi \int_0^{\pi/6} \cos 2x \sqrt{1 + 4 \sin^2 2x} dx. \end{aligned}$$

Let $u = 2 \sin 2x$, so $du = 4 \cos 2x dx$. Then

$$\begin{aligned} S &= \frac{\pi}{2} \int_0^{\sqrt{3}} \sqrt{1 + u^2} du. \\ &= \frac{\pi}{2} \left[\frac{1}{2} u \sqrt{1 + u^2} + \frac{1}{2} \ln(u + \sqrt{1 + u^2}) \right]_0^{\sqrt{3}}. \\ &= \frac{\pi\sqrt{3}}{2} + \frac{\pi}{4} \ln(2 + \sqrt{3}). \end{aligned}$$

Question 10

Find the area of the surface obtained by rotating the curve about the x -axis.

$$y = \cosh x, \quad 0 \leq x \leq 1.$$

Solution

$$\begin{aligned} \frac{dy}{dx} &= \sinh x, & 1 + \sinh^2 x &= \cosh^2 x. \\ S &= 2\pi \int_0^1 \cosh x \cdot \cosh x dx = 2\pi \int_0^1 \cosh^2 x dx. \\ &= 2\pi \int_0^1 \frac{1}{2} (1 + \cosh 2x) dx = \pi \left[x + \frac{1}{2} \sinh 2x \right]_0^1. \\ &= \pi \left(1 + \frac{1}{2} \sinh 2 \right). \end{aligned}$$

Improper Integral

Question 1

$$\int_1^{\infty} \frac{1}{(3x+1)^2} dx$$

Solution

$$= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{(3x+1)^2} dx.$$

Now

$$\begin{aligned} \int \frac{1}{(3x+1)^2} dx &= \frac{1}{3} \int \frac{1}{u^2} du \quad [u = 3x+1, \quad du = 3 dx] \\ &= -\frac{1}{3u} + C = -\frac{1}{3(3x+1)} + C. \end{aligned}$$

So

$$I = \lim_{t \rightarrow \infty} \left[-\frac{1}{3(3x+1)} \right]_1^t = \lim_{t \rightarrow \infty} \left(-\frac{1}{3(3t+1)} + \frac{1}{12} \right) = 0 + \frac{1}{12} = \frac{1}{12}.$$

Convergent.

Question 2

$$\int_{-\infty}^0 \frac{1}{2x-5} dx$$

Solution

$$\begin{aligned} &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{2x-5} dx = \lim_{t \rightarrow -\infty} \left[\frac{1}{2} \ln |2x-5| \right]_t^0 \\ &= \lim_{t \rightarrow -\infty} \left(\frac{1}{2} \ln 5 - \frac{1}{2} \ln |2t-5| \right) = -\infty. \end{aligned}$$

Divergent.

Question 3

$$\int_{-\infty}^{-1} \frac{1}{\sqrt{2-w}} dw$$

Solution

$$\begin{aligned} &= \lim_{t \rightarrow -\infty} \int_t^{-1} \frac{1}{\sqrt{2-w}} dw = \lim_{t \rightarrow -\infty} \left[-2\sqrt{2-w} \right]_t^{-1} \\ &= \lim_{t \rightarrow -\infty} \left[-2\sqrt{3} + 2\sqrt{2-t} \right] = \infty. \end{aligned}$$

Divergent.

Question 4

$$\int_0^{\infty} \frac{x}{(x^2+2)^2} dx$$

Solution

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \int_0^t \frac{x}{(x^2+2)^2} dx \\ &= \lim_{t \rightarrow \infty} \frac{1}{2} \int_2^{t^2+2} \frac{1}{u^2} du = \frac{1}{2} \lim_{t \rightarrow \infty} \left[-\frac{1}{u} \right]_2^{t^2+2} \\ &= \frac{1}{2} \lim_{t \rightarrow \infty} \left(-\frac{1}{t^2+2} + \frac{1}{2} \right) = \frac{1}{2} \left(0 + \frac{1}{2} \right) = \frac{1}{4}. \end{aligned}$$

Convergent.

Question 5

$$\int_4^{\infty} e^{-y/2} dy$$

Solution

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \int_4^t e^{-y/2} dy = \lim_{t \rightarrow \infty} \left[-2e^{-y/2} \right]_4^t \\ &= \lim_{t \rightarrow \infty} \left(-2e^{-t/2} + 2e^{-2} \right) = 2e^{-2}. \end{aligned}$$

Convergent.

Question 6

$$\int_{-\infty}^{-1} e^{-2t} dt$$

Solution

$$\begin{aligned} &= \lim_{x \rightarrow -\infty} \int_x^{-1} e^{-2t} dt = \lim_{x \rightarrow -\infty} \left[-\frac{1}{2} e^{-2t} \right]_x^{-1} \\ &= \lim_{x \rightarrow -\infty} \left(-\frac{1}{2} e^2 + \frac{1}{2} e^{-2x} \right) = \infty. \end{aligned}$$

Divergent.

Question 7

$$\int_{-\infty}^{\infty} \frac{x}{1+x^2} dx$$

Solution

$$= \int_{-\infty}^0 \frac{x}{1+x^2} dx + \int_0^{\infty} \frac{x}{1+x^2} dx.$$

Let

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2).$$

Thus

$$\int_0^{\infty} \frac{x}{1+x^2} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln(1+x^2) \right]_0^t = \infty,$$

and

$$\int_{-\infty}^0 \frac{x}{1+x^2} dx = \lim_{t \rightarrow -\infty} \left[\frac{1}{2} \ln(1+x^2) \right]_t^0 = -\infty.$$

Therefore the integral is divergent.

Question 8

$$\int_{-\infty}^{\infty} (2-v^4) dv$$

Solution

$$= \int_{-\infty}^0 (2-v^4) dv + \int_0^{\infty} (2-v^4) dv.$$

Now

$$\int (2-v^4) dv = 2v - \frac{1}{5}v^5.$$

Thus

$$\int_0^{\infty} (2-v^4) dv = \lim_{t \rightarrow \infty} \left[2v - \frac{1}{5}v^5 \right]_0^t = -\infty,$$

and

$$\int_{-\infty}^0 (2 - v^4) dv = \lim_{t \rightarrow -\infty} \left[2v - \frac{1}{5}v^5 \right]_t^0 = \infty.$$

Since both integrals diverge, the given integral diverges.

Question 9

$$\int_{-\infty}^{\infty} x e^{-x^2} dx$$

Solution

$$= \int_{-\infty}^0 x e^{-x^2} dx + \int_0^{\infty} x e^{-x^2} dx.$$

Let

$$\int x e^{-x^2} dx = -\frac{1}{2} e^{-x^2}.$$

Then

$$\int_0^{\infty} x e^{-x^2} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{2} e^{-x^2} \right]_0^t = \frac{1}{2},$$

and

$$\int_{-\infty}^0 x e^{-x^2} dx = \lim_{t \rightarrow -\infty} \left[-\frac{1}{2} e^{-x^2} \right]_t^0 = -\frac{1}{2}.$$

Therefore,

$$\int_{-\infty}^{\infty} x e^{-x^2} dx = 0.$$

Convergent.

Question 10

$$\int_{-\infty}^{\infty} x^2 e^{-x^3} dx$$

Solution

$$= \int_{-\infty}^0 x^2 e^{-x^3} dx + \int_0^{\infty} x^2 e^{-x^3} dx.$$

Let

$$\int x^2 e^{-x^3} dx = -\frac{1}{3} e^{-x^3}.$$

Then

$$\int_0^{\infty} x^2 e^{-x^3} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{3} e^{-x^3} \right]_0^t = \frac{1}{3},$$

but

$$\int_{-\infty}^0 x^2 e^{-x^3} dx = \lim_{t \rightarrow -\infty} \left[-\frac{1}{3} e^{-x^3} \right]_t^0 = \infty.$$

Hence the integral diverges.

Question 11

$$\int_{2\pi}^{\infty} \sin \theta d\theta$$

Solution

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \int_{2\pi}^t \sin \theta d\theta = \lim_{t \rightarrow \infty} [-\cos \theta]_{2\pi}^t \\ &= \lim_{t \rightarrow \infty} (-\cos t + \cos 2\pi). \end{aligned}$$

Since $\cos t$ does not approach a limit as $t \rightarrow \infty$, the given integral diverges. Divergent.

Question 12

$$\int_0^{\infty} \cos^2 \alpha \, d\alpha$$

Solution Using the identity $\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$,

$$\int \cos^2 \alpha \, d\alpha = \frac{1}{2} \int (1 + \cos 2\alpha) \, d\alpha = \frac{1}{2} \left(\alpha + \frac{1}{2} \sin 2\alpha \right).$$

Thus

$$\int_0^{\infty} \cos^2 \alpha \, d\alpha = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \left(\alpha + \frac{1}{2} \sin 2\alpha \right) \right]_0^t = \infty.$$

Divergent.

Question 13

$$\int_1^{\infty} \frac{x+1}{x^2+2x} \, dx$$

Solution

$$= \int_1^{\infty} \frac{x+1}{x(x+2)} \, dx = \frac{1}{2} \int_1^{\infty} \frac{2x+2}{x(x+2)} \, dx.$$

Using partial fractions,

$$\frac{2x+2}{x(x+2)} = \frac{1}{x} + \frac{1}{x+2}.$$

Thus

$$\begin{aligned} \int_1^{\infty} \frac{x+1}{x^2+2x} \, dx &= \frac{1}{2} \int_1^{\infty} \left(\frac{1}{x} + \frac{1}{x+2} \right) \, dx \\ &= \frac{1}{2} [\ln x + \ln(x+2)]_1^t = \frac{1}{2} \ln(t(t+2)) - \frac{1}{2} \ln 3. \end{aligned}$$

Since $\ln(t(t+2)) \rightarrow \infty$ as $t \rightarrow \infty$, the integral diverges. Divergent.

Question 14

$$\int_0^{\infty} \frac{dz}{z^2+3z+2}$$

Solution

$$= \int_0^{\infty} \frac{1}{(z+1)(z+2)} \, dz = \int_0^{\infty} \left(\frac{1}{z+1} - \frac{1}{z+2} \right) \, dz.$$

Thus

$$\begin{aligned} &= \lim_{t \rightarrow \infty} [\ln(z+1) - \ln(z+2)]_0^t \\ &= \lim_{t \rightarrow \infty} \ln \frac{t+1}{t+2} - \ln \frac{1}{2} = \ln 2. \end{aligned}$$

Convergent.

Question 15

$$\int_0^{\infty} s e^{-5s} \, ds$$

Solution Using integration by parts, let $u = s$, $dv = e^{-5s} ds$. Then $du = ds$, $v = -\frac{1}{5}e^{-5s}$.

$$\int s e^{-5s} \, ds = -\frac{1}{5} s e^{-5s} + \frac{1}{5} \int e^{-5s} \, ds = -\frac{1}{5} s e^{-5s} - \frac{1}{25} e^{-5s}.$$

Thus

$$\int_0^{\infty} s e^{-5s} \, ds = \lim_{t \rightarrow \infty} \left[-\frac{1}{5} s e^{-5s} - \frac{1}{25} e^{-5s} \right]_0^t = \frac{1}{25}.$$

Convergent.

Question 16

$$\int_{-\infty}^6 re^{r/3} dr$$

Solution Using integration by parts, let $u = r$, $dv = e^{r/3} dr$. Then $du = dr$, $v = 3e^{r/3}$.

$$\int re^{r/3} dr = 3re^{r/3} - 3 \int e^{r/3} dr = 3re^{r/3} - 9e^{r/3}.$$

Thus

$$\begin{aligned}\int_{-\infty}^6 re^{r/3} dr &= \lim_{t \rightarrow -\infty} \left[3re^{r/3} - 9e^{r/3} \right]_t^6 \\ &= 9e^2.\end{aligned}$$

Convergent.

Question 17

$$\int_1^{\infty} \frac{\ln x}{x} dx$$

Solution

$$= \lim_{t \rightarrow \infty} \int_1^t \frac{\ln x}{x} dx.$$

Let $u = \ln x$, so $du = \frac{1}{x} dx$. Then

$$\int \frac{\ln x}{x} dx = \frac{1}{2}(\ln x)^2.$$

Thus

$$\int_1^{\infty} \frac{\ln x}{x} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{2}(\ln x)^2 \right]_1^t = \infty.$$

Divergent.

Question 18

$$\int_{-\infty}^{\infty} e^{-|x|} dx$$

Solution

$$= \int_{-\infty}^0 e^x dx + \int_0^{\infty} e^{-x} dx.$$

Now

$$\int_{-\infty}^0 e^x dx = \lim_{t \rightarrow -\infty} [e^x]_t^0 = 1,$$

and

$$\int_0^{\infty} e^{-x} dx = \lim_{t \rightarrow \infty} [-e^{-x}]_0^t = 1.$$

Therefore,

$$\int_{-\infty}^{\infty} e^{-|x|} dx = 2.$$

Convergent.

Question 19

$$\int_{-\infty}^{\infty} \frac{x^2}{9+x^6} dx$$

Solution

$$= 2 \int_0^{\infty} \frac{x^2}{9+x^6} dx.$$

Let $u = x^3$, so $du = 3x^2 dx$. Then

$$\int \frac{x^2}{9+x^6} dx = \frac{1}{3} \int \frac{1}{9+u^2} du = \frac{1}{9} \arctan \frac{u}{3}.$$

Thus

$$\int_{-\infty}^{\infty} \frac{x^2}{9+x^6} dx = 2 \left[\frac{1}{9} \arctan \frac{x^3}{3} \right]_0^{\infty} = \frac{2}{9} \cdot \frac{\pi}{2} = \frac{\pi}{9}.$$

Convergent.

Question 20

$$\int_1^{\infty} \frac{\ln x}{x^3} dx$$

Solution Using integration by parts, let $u = \ln x$, $dv = x^{-3} dx$. Then $du = \frac{1}{x} dx$, $v = -\frac{1}{2x^2}$.

$$\int \frac{\ln x}{x^3} dx = -\frac{\ln x}{2x^2} + \frac{1}{2} \int \frac{1}{x^3} dx = -\frac{\ln x}{2x^2} - \frac{1}{4x^2}.$$

Thus

$$\int_1^{\infty} \frac{\ln x}{x^3} dx = \lim_{t \rightarrow \infty} \left[-\frac{\ln x}{2x^2} - \frac{1}{4x^2} \right]_1^t = \frac{1}{4}.$$

Convergent.

Question 21

$$\int_1^{\infty} \frac{\ln x}{x^2} dx$$

Solution Using integration by parts, let $u = \ln x$, $dv = x^{-2} dx$. Then $du = \frac{1}{x} dx$, $v = -\frac{1}{x}$.

$$\int \frac{\ln x}{x^2} dx = -\frac{\ln x}{x} + \int \frac{1}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x}.$$

Thus

$$\int_1^{\infty} \frac{\ln x}{x^2} dx = \lim_{t \rightarrow \infty} \left[-\frac{\ln x}{x} - \frac{1}{x} \right]_1^t = 1.$$

Convergent.

Question 22

$$\int_0^{\infty} \frac{x \arctan x}{(1+x^2)^2} dx$$

Solution Using integration by parts, let $u = \arctan x$, $dv = \frac{x}{(1+x^2)^2} dx$. Then $du = \frac{1}{1+x^2} dx$, $v = -\frac{1}{2(1+x^2)}$.

$$\int \frac{x \arctan x}{(1+x^2)^2} dx = -\frac{\arctan x}{2(1+x^2)} + \frac{1}{2} \int \frac{1}{1+x^2} dx.$$

Thus

$$\int_0^{\infty} \frac{x \arctan x}{(1+x^2)^2} dx = \left[-\frac{\arctan x}{2(1+x^2)} \right]_0^{\infty} + \frac{1}{2} [\arctan x]_0^{\infty} = \frac{\pi}{8}.$$

Convergent.

Question 23

$$\int_0^3 \frac{1}{\sqrt{x}} dx$$

Solution

$$= \lim_{t \rightarrow 0^+} \int_t^3 x^{-1/2} dx = \lim_{t \rightarrow 0^+} [2\sqrt{x}]_t^3 = 2\sqrt{3}.$$

Convergent.

Question 24

$$\int_0^3 \frac{1}{x\sqrt{x}} dx$$

Solution

$$= \lim_{t \rightarrow 0^+} \int_t^3 x^{-3/2} dx = \lim_{t \rightarrow 0^+} [-2x^{-1/2}]_t^3 = \infty.$$

Divergent.

Question 25

$$\int_{-1}^0 \frac{1}{x^2} dx$$

Solution

$$= \lim_{t \rightarrow 0^-} \int_{-1}^t x^{-2} dx = \lim_{t \rightarrow 0^-} \left[-\frac{1}{x} \right]_{-1}^t = \infty.$$

Divergent.

Question 26

$$\int_1^9 \frac{1}{\sqrt[3]{x-9}} dx$$

Solution

$$\begin{aligned} &= \lim_{t \rightarrow 9^-} \int_1^t (x-9)^{-1/3} dx = \lim_{t \rightarrow 9^-} \left[\frac{3}{2}(x-9)^{2/3} \right]_1^t \\ &= \frac{3}{2}(0 - (-8)^{2/3}) = 6. \end{aligned}$$

Convergent.

Question 27

Use the Comparison Theorem to determine whether the integral is convergent or divergent.

Question 28

$$\int_1^\infty \frac{\cos^2 x}{1+x^2} dx$$

Solution For $x \geq 1$,

$$0 \leq \frac{\cos^2 x}{1+x^2} \leq \frac{1}{1+x^2} \leq \frac{1}{x^2}.$$

But

$$\int_1^\infty \frac{1}{x^2} dx$$

is convergent by Equation 2 with $p = 2 > 1$. Therefore,

$$\int_1^\infty \frac{\cos^2 x}{1+x^2} dx$$

is convergent by the Comparison Theorem.

Question 29

$$\int_1^{\infty} \frac{2 + e^{-x}}{x} dx$$

Solution For $x \geq 1$,

$$\frac{2 + e^{-x}}{x} \geq \frac{2}{x},$$

since $e^{-x} > 0$. But

$$\int_1^{\infty} \frac{1}{x} dx$$

is divergent by Equation 2 with $p = 1 \leq 1$. Therefore,

$$\int_1^{\infty} \frac{2 + e^{-x}}{x} dx$$

is divergent by the Comparison Theorem.

Question 30

$$\int_1^{\infty} \frac{dx}{x + e^{2x}}$$

Solution For $x \geq 1$,

$$x + e^{2x} \geq e^{2x}, \quad \text{so} \quad \frac{1}{x + e^{2x}} \leq e^{-2x}.$$

Now

$$\int_1^{\infty} e^{-2x} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{2} e^{-2x} \right]_1^t = \frac{1}{2} e^{-2},$$

which is convergent. Therefore,

$$\int_1^{\infty} \frac{dx}{x + e^{2x}}$$

is convergent by the Comparison Theorem.

Question 31

$$\int_1^{\infty} \frac{x}{\sqrt{1 + x^6}} dx$$

Solution For $x \geq 1$,

$$0 < \frac{x}{\sqrt{1 + x^6}} \leq \frac{x}{\sqrt{x^6}} = \frac{1}{x^2}.$$

But

$$\int_1^{\infty} \frac{1}{x^2} dx$$

is convergent by Equation 2 with $p = 2 > 1$. Therefore,

$$\int_1^{\infty} \frac{x}{\sqrt{1 + x^6}} dx$$

is convergent by the Comparison Theorem.

Question 32

$$\int_0^{\pi/2} \frac{dx}{x \sin x}$$

Solution Since $0 \leq \sin x \leq x$ on $(0, \pi/2]$, we have

$$\frac{1}{x \sin x} \geq \frac{1}{x^2}.$$

But

$$\int_0^{\pi/2} \frac{1}{x^2} dx$$

is divergent. Therefore,

$$\int_0^{\pi/2} \frac{dx}{x \sin x}$$

is divergent by the Comparison Theorem.

Question 33

$$\int_0^1 \frac{e^{-x}}{\sqrt{x}} dx$$

Solution For $0 \leq x \leq 1$,

$$e^{-x} \leq 1, \quad \text{so} \quad \frac{e^{-x}}{\sqrt{x}} \leq \frac{1}{\sqrt{x}}.$$

Now

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} dx = 2,$$

which is convergent. Therefore,

$$\int_0^1 \frac{e^{-x}}{\sqrt{x}} dx$$

is convergent by the Comparison Theorem.